

Fig. 3.2 npn-transistor in CB configuration

The emitter-base junction is equivalent to a p - n junction diode. Hence at the input, the common base transistor can be represented by a p - n diode.

In common-base configuration the current gain alpha is given by

$$\alpha = \frac{I_c}{I_e}$$

or

$$I_c = \alpha I_e \quad (3.5)$$

Note that the collector current I_c is controlled by the emitter current I_e . Hence at the output the common-base transistor can be represented by a controlled current source.

Since I_c flows through the collector-base junction, the controlled current source appears between the collector and base terminals.

The r_e model of n pn transistor in CB configuration is shown in Fig. 3.3.

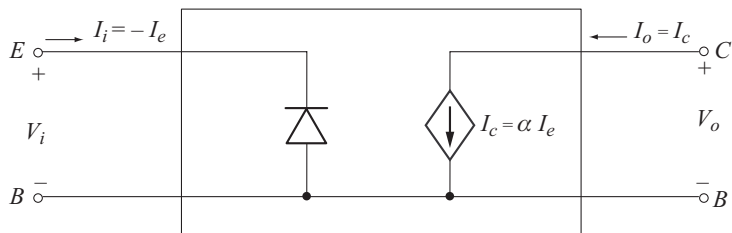


Fig. 3.3 r_e model of transistor in CB configuration

For ac response, the diode can be replaced by its equivalent ac resistance, r_e . The ac resistance of emitter-base diode is given in Equation (3.2) as

$$r_e = \frac{26\text{mV}}{I_E} \quad (3.6)$$

The common-base r_e equivalent circuit is obtained by replacing the diode in Fig. 3.3 by its ac resistance r_e . This is shown in Fig. 3.4.

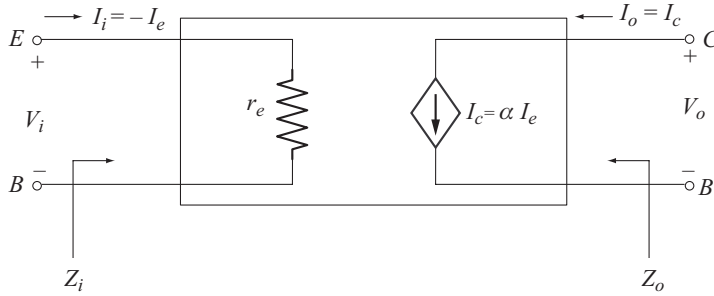


Fig. 3.4 Common base r_e equivalent circuit

From the circuit of Fig. 3.4, let us find the following performance parameters :

- Input impedance Z_i is given by

$$Z_i = \frac{V_i}{I_i} = r_e \quad (3.7)$$

Note that the input impedance is quite low, which is in the order of few tens of ohms.

- Output impedance $Z_o = \infty \Omega$, since the output circuit contains an ideal current source. The output impedance is actually given by the slope of output characteristics. Typically it is in the range of 1–2 M Ω which is quite a high value.
- With R_L connected between the output terminals, the voltage gain is given by

$$A_v = \frac{V_o}{V_i} = \frac{\alpha R_L}{r_e} \quad (3.8)$$

Since A_v is positive, V_o and V_i are in phase. Also, due to low value of r_e , A_v is quite high.

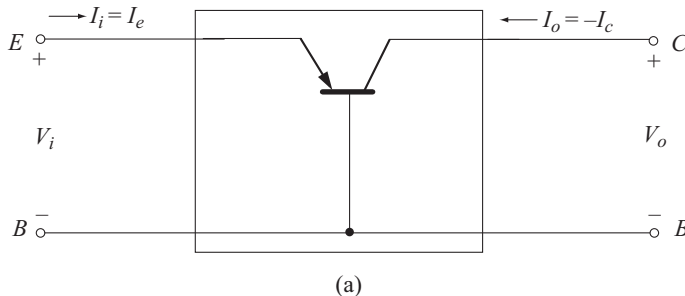
- With R_L connected between the output terminals, the current gain is given by

$$A_i = \frac{I_o}{I_i} = -\alpha \approx -1 \quad (3.9)$$

Note that the magnitude of current gain is approximately unity.

◆ 3.6 r_e MODEL OF PNP TRANSISTOR IN CB CONFIGURATION

Figure 3.5 shows the r_e model of pnp transistor in CB configuration. Note that for pnp transistor, I_e is into the transistor. Therefore I_b and I_c are out of the transistor.



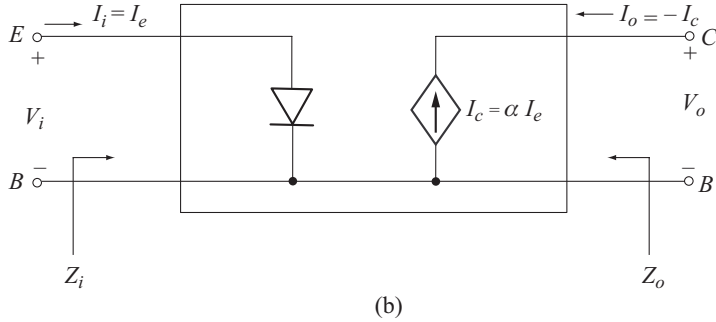


Fig. 3.5 r_e model of pnp transistor in CB configuration

It can be verified that, all results obtained for CB amplifier using npn transistor can be applied for this case also. Thus ac analysis of transistor amplifier is independent of the type of transistor (whether npn or pnp) used in the circuit.

Example 3.1

For the common base configuration of Fig. 3.2, $I_E = 4 \text{ mA}$, $\alpha = 0.991$. An ac signal of 3 mV is applied between the base and emitter terminals. If $R_L = 610 \Omega$. Calculate

- | | |
|-----------------------------|---|
| (a) r_e and Z_i | (b) A_V and A_I |
| (c) V_o , I_i and I_o | (d) Z_o with $r_o = \infty$ and I_c , I_e and I_b |

Solution

$$(a) \quad r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{4 \text{ mA}} = 6.5 \Omega$$

$$Z_i = r_e = 6.5 \Omega$$

$$(b) \quad A_V = \frac{\alpha R_L}{r_e} = \frac{(0.991)(610 \Omega)}{(6.5 \Omega)} = 93$$

$$A_I = -\alpha = -0.991$$

$$(c) \quad V_o = A_V V_i = (93)(3 \text{ mV}) = 279 \text{ mV}$$

$$I_i = \frac{V_i}{Z_i} = \frac{3 \text{ mV}}{6.5 \Omega} = 461.52 \mu\text{A}$$

$$I_o = A_I I_i = -(0.991)(461.52 \mu\text{A}) = -457.36 \mu\text{A}$$

$$(d) \quad Z_o = r_o = \infty$$

$$I_i = -I_e \quad \text{and} \quad I_o = I_c$$

To avoid confusion with signs, let us take only the magnitudes of I_c , I_b and I_e .

$$\therefore I_e = 461.52 \mu\text{A}$$

$$I_c = 457.36 \mu\text{A}$$

$$\text{Also } I_e = I_b + I_c$$

$$\therefore I_b = I_e - I_c = 461.52 \mu\text{A} - 457.36 \mu\text{A} = 4.16 \mu\text{A}$$

◆ 3.7 THE r_e MODEL OF TRANSISTOR IN COMMON-EMITTER (CE) CONFIGURATION

Figure 3.6 shows an npn transistor in CE configuration represented as a two port network. The input terminals are the base and emitter terminals, and the output terminals are the collector and emitter terminals. Note that the emitter terminal is common between the input and output ports. The input current is I_b and the output current is I_c .

$$\therefore I_i = I_b \quad (3.10)$$

$$\text{and } I_o = I_c \quad (3.11)$$

The emitter-base junction is equivalent to a p - n junction diode. Hence between the emitter and base terminals, the common-emitter transistor can be modelled as a p - n diode.

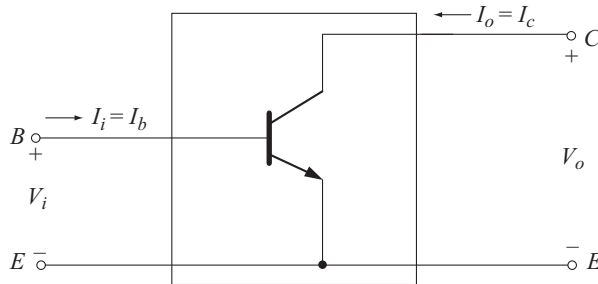


Fig. 3.6 npn transistor in CE configuration

In common-emitter configuration the current gain Beta is given by

$$\beta = \frac{I_c}{I_b}$$

$$\text{or } I_c = \beta I_b \quad (3.12)$$

Note that the collector current I_c is controlled by the base current I_b . Since I_c flows through the collector-base junction, the controlled current source is connected between the collector and base terminals. The r_e model is shown in Fig. 3.7.

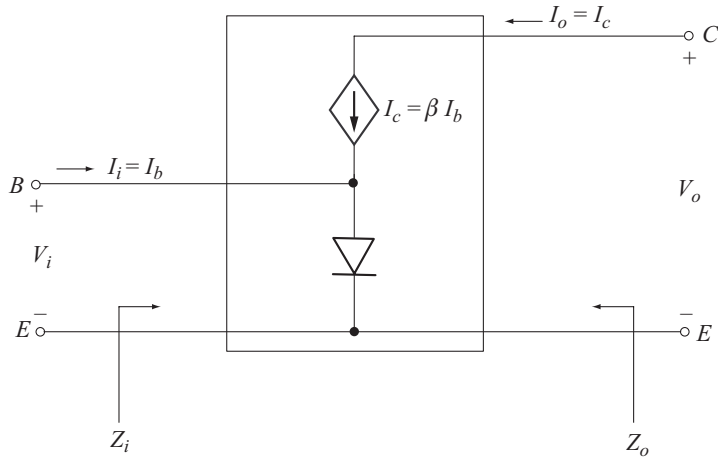


Fig. 3.7 r_e model of transistor in CE Configuration

For ac response, the diode can be replaced by its equivalent ac resistance, r_e as shown in Fig. 3.8.

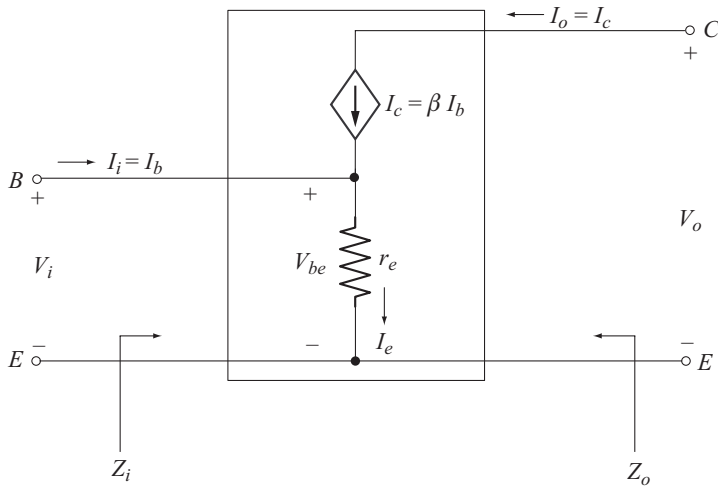


Fig. 3.8 Common emitter r_e equivalent circuit

Input Impedance (Z_i)

$$Z_i = \frac{V_i}{I_i} \quad (3.13)$$

$$\begin{aligned} V_i &= V_{be} = I_e r_e = (1 + \beta) I_b r_e \\ &\approx \beta I_b r_e \quad (\text{since } 1 + \beta \approx \beta) \end{aligned} \quad (3.14)$$

Also

$$I_i = I_b$$

$$\begin{aligned} \text{Now} \quad Z_i &= \frac{\beta I_b r_e}{I_b} \\ \text{or} \quad Z_i &= \beta r_e \end{aligned} \quad (3.15)$$

Note that the input impedance of transistor in CE configuration is β times the value of r_e . Stated alternatively, **a resistive element in the emitter leg is reflected into the input circuit with a multiplying factor β** .

For instance if $r_e = 6.5 \Omega$ and $\beta = 200$, the input impedance is

$$Z_i = \beta r_e = (200)(6.5 \Omega) = 1.3 \text{ k}\Omega$$

For the common-emitter configuration, typical values of Z_i range from a few hundred ohms to the kilo ohms range, with a maximum of about 6 k Ω to 7 k Ω .

Output Impedance (Z_o)

The output impedance is given by the reciprocal of output characteristic. Typically it is in the range of 40–50 k Ω . Z_o is also denoted by r_o .

Voltage Gain (A_v) and Current Gain (A_i)

With R_L connected between the output terminals, the voltage gain is given by

$$A_v = -\frac{R_L}{r_e} \quad (3.16)$$

The negative sign implies that V_o and V_i are 180° out of phase. The magnitude of voltage gain is

$$|A_v| = \frac{R_L}{r_e} \quad (3.17)$$

$$\text{Also the current gain is} \quad A_i = \beta \quad (3.18)$$

Observe that due to low value of r_e and large value of β , the voltage and current gains are quite high.

◆ 3.8 r_e MODEL OF TRANSISTOR IN CE CONFIGURATION INCLUDING r_o

In CE configuration:

- The input impedance is measured between the base and emitter terminals. Thus a resistance βr_e is connected between the base and emitter terminals.
- The output resistance is measured between the collector and emitter terminals. Thus a resistance r_o is connected between the collector and emitter terminals.
- The emitter current I_e is given by

$$\begin{aligned} I_e &= I_b + I_c \\ &= I_b + \beta I_b \end{aligned}$$

Note that the contribution of collector current to make emitter current is βI_b .

Thus a controlled current source of current βI_b is connected between the collector and emitter terminals.

Figure 3.9 shows the r_e model of transistor in CE configuration including the effect of r_o .

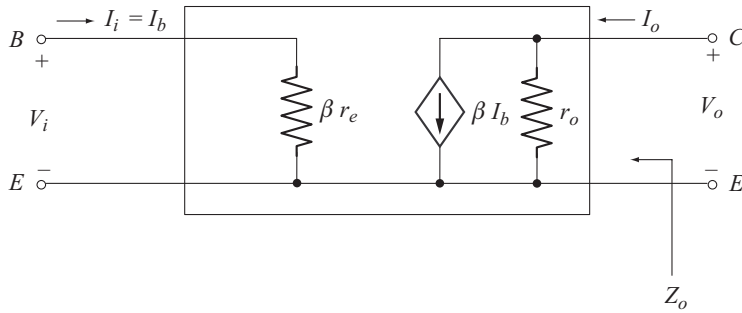


Fig. 3.9 r_e model of CE configuration including r_o

Note: For common-collector (CC) configuration, the model defined for the CE configuration of Fig. 3.8 is usually employed instead of defining a separate model for the CC configuration.

◆ 3.9 THE HYBRID EQUIVALENT MODEL

Consider the two port network shown in Fig. 3.10. To develop the hybrid equivalent model of this two port network, let us select I_i and V_o as independent variables and V_i and I_o as dependent variables.

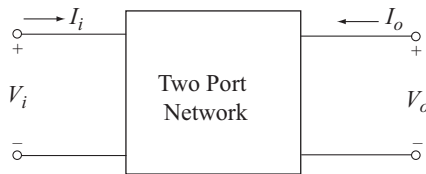


Fig. 3.10 Two port network

We can express V_i and I_o in terms of I_i and V_o as follows:

$$V_i = f_1(I_i, V_o) \quad (3.19)$$

$$I_o = f_2(I_i, V_o) \quad (3.20)$$

Note that we are developing a small signal low frequency model which is a linear model. Thus we can express V_i and I_o as a linear combinations of I_i and V_o .

$$\therefore V_i = h_{11} I_i + h_{12} V_o \quad (3.21)$$

$$\text{and } I_o = h_{21} I_i + h_{22} V_o \quad (3.22)$$

The parameters h_{11} , h_{12} , h_{21} and h_{22} are called the hybrid parameters or h -parameters. The name hybrid is due to the fact that they do not have the same units. Let us now, define these parameters.

Set $V_o = 0$ by short circuiting the output terminals:

From Equation (3.21),

$$h_{11} = \left. \frac{V_i}{I_i} \right|_{V_o=0} \quad (3.23)$$

h_{11} is called the input impedance with output short circuited and has the unit ohms.

From Equation (3.22),

$$h_{21} = \left. \frac{I_o}{I_i} \right|_{V_o=0} \quad (3.24)$$

h_{21} is the short circuit forward transfer current ratio and has no unit

Set $I_i = 0$ by open circuiting the input:

From Equation (3.21),

$$h_{12} = \left. \frac{V_i}{V_o} \right|_{I_i=0} \quad (3.25)$$

h_{12} is the open circuit reverse transfer voltage ratio and has no unit

From Equation (3.22),

$$h_{22} = \left. \frac{I_o}{V_o} \right|_{I_i=0} \quad (3.26)$$

h_{22} is the open circuit output admittance and has the unit mhos.

Note that:

h_{11} has the unit ohms

h_{22} has the unit mhos

h_{12} and h_{21} have no units.

Since these parameters dimensionally differ, they are called hybrid parameters. The double subscript parameter notation can further be reduced to a single subscript notation as follows:

Let i represent 11 denoting input

o represent 22 denoting output

f represent 21 denoting forward transfer

and r represent 12 denoting reverse transfer

Equations (3.21) and (3.22) now become

$$V_i = h_i I_i + h_r V_o \quad (3.27)$$

$$I_o = h_f I_i + h_o V_o \quad (3.28)$$

The circuit model based on Equation (3.27) is shown in Fig. 3.11 and Fig. 3.12 shows the circuit model of Equation (3.28). Note that Equation (3.27) is a KVL equation and Equation (3.28) is a KCL equation.

Equation (3.27) has two components:

- $h_i I_i$ representing a voltage drop across the impedance h_i and
- $h_r V_o$ representing a controlled voltage source.

Equation (3.28) has two components:

- $h_f I_i$ representing a controlled current source and
- $h_o V_o$ representing the current through admittance h_o .

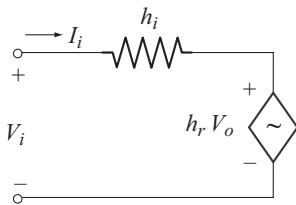


Fig. 3.11 Circuit model of $V_i = h_i I_i + h_r V_o$

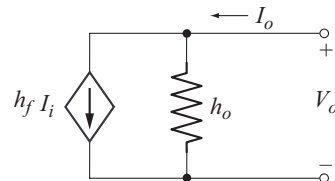


Fig. 3.12 Circuit model of $I_o = h_f I_i + h_o V_o$

Combining these two models, we obtain the hybrid model of the two-port network as shown in Fig. 3.13.

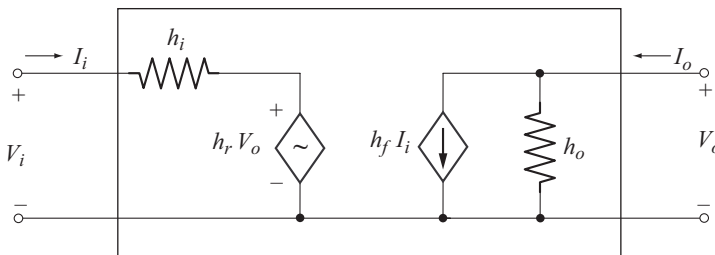


Fig. 3.13 Complete hybrid equivalent circuit

3.9.1 Hybrid Parameter Nomenclature for Transistor

In order to specify the hybrid parameters for a given transistor configuration, a second subscript is used:

e for CE Configuration

b for CB Configuration

and c for CC Configuration

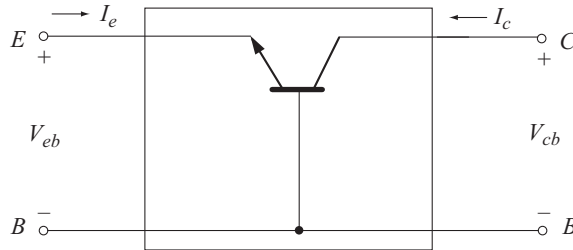
This is illustrated in Table 3.1.

Table 3.1 Hybrid parameter nomenclature

Hybrid Parameters	Configuration		
	CE	CB	CC
h_i	h_{ie}	h_{ib}	h_{ic}
h_f	h_{fe}	h_{fb}	h_{fc}
h_r	h_{re}	h_{rb}	h_{rc}
h_o	h_{oe}	h_{ob}	h_{oc}

◆ 3.10 HYBRID MODEL OF CB CONFIGURATION

The two port network representation of CB configuration is shown in Fig. 3.14.

**Fig. 3.14** Two port network representation of CB configuration

Comparing the network of Fig. 3.10 and Fig. 3.14, we can list the equivalent voltages and currents as given in Table 3.2.

Table 3.2 Equivalent voltages and currents for CB configuration

General two port network	CB configuration
V_i	V_{eb}
I_i	I_e
V_o	V_{cb}
I_o	I_c

Equations (3.21) and (3.22) now become

$$V_{eb} = h_{ib} I_e + h_{rb} V_{cb} \quad (3.29)$$

$$I_c = h_{fb} I_e + h_{ob} V_{cb} \quad (3.30)$$

The hybrid model constructed based on Equations (3.29) and (3.30) is shown in Fig. 3.15.

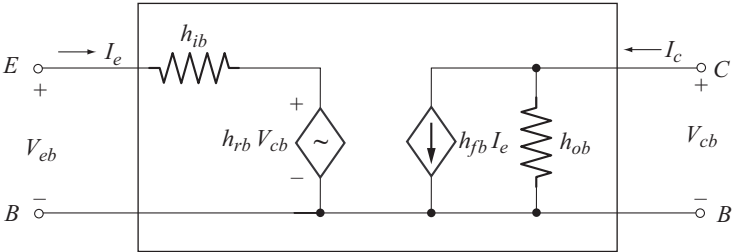


Fig. 3.15 Hybrid model of CB configuration

◆ 3.11 HYBRID MODEL OF CE CONFIGURATION

The two port network representation of CE configuration is shown in Fig. 3.16.

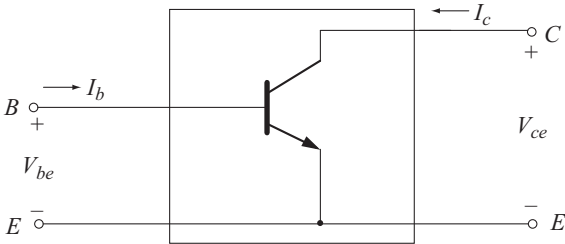


Fig. 3.16 Two port network representation of CE configuration

Comparing the network of Fig. 3.10 and Fig. 3.16 we can list the equivalent voltages and currents as given in Table.3.3.

Table 3.3 Equivalent voltages and currents for CE configuration

General two port network	CE Configuration
V_i	V_{be}
I_i	I_b
V_o	V_{ce}
I_o	I_c

Equations (3.21) and (3.22) now become

$$V_{be} = h_{ie} I_b + h_{re} V_{ce} \tag{3.31}$$

$$I_c = h_{fe} I_b + h_{oe} V_{ce} \tag{3.32}$$

Based on these equations, the hybrid model for the CE configuration can be developed as shown in Fig. 3.17.

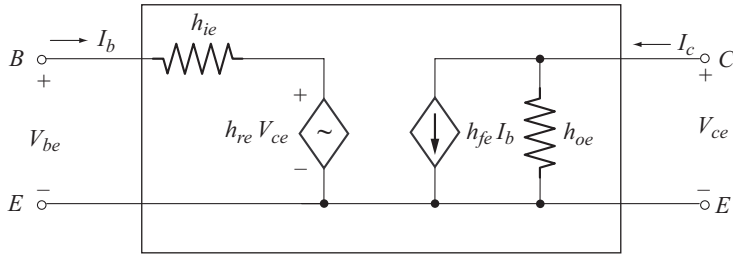


Fig. 3.17 Hybrid model of CE configuration

◆ 3.12 HYBRID MODEL OF CC CONFIGURATION

The two port network representation of CC configuration is shown in Fig. 3.18.

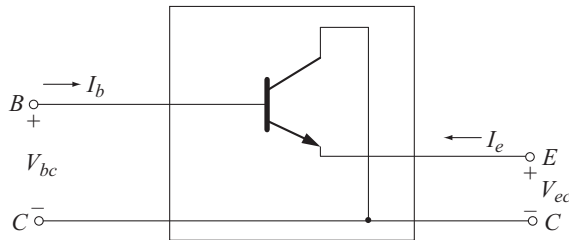


Fig. 3.18 Two port network representation of CC configuration

Comparing the networks of Fig. 3.10 and 3.18 we can list the equivalent voltages and currents as given in Table 3.4.

Table 3.4 Equivalent voltages and currents for CC configuration

General two port network	CC configuration
V_i	V_{bc}
I_i	I_b
V_o	V_{ec}
I_o	I_e

Equations (3.21) and (3.22) now become

$$V_{bc} = h_{ic} I_b + h_{rc} V_{ec} \quad (3.33)$$

$$I_e = h_{fc} I_b + h_{oc} V_{ec} \quad (3.34)$$

Based on these equations, the hybrid model for the CC configuration can be developed as shown in Fig. 3.19.

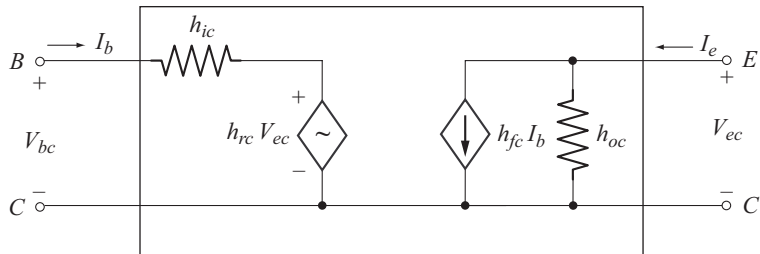


Fig. 3.19 Hybrid model of CC configuration

Typical values of hybrid parameters for all the three configurations are given in Table 3.5.

Table 3.5 Typical values of h -parameters in CE, CB and CC configurations

Parameter	Configuration		
	CE	CB	CC
h_i	1 k Ω	20 Ω	1 k Ω
h_r	2.5×10^{-4}	3.0×10^{-4}	1
h_f	50	-0.98	-50
h_o	25 μ A/V	0.5 μ A/V	25 μ A/V
$1/h_o$	40 k Ω	2 M Ω	40 k Ω

◆ 3.13 VARIATIONS OF h -PARAMETERS

The h -parameters in general vary with changes in temperature, frequency, voltage and current. It is worthwhile to look at the variation of h -parameters particularly with reference to collector current, collector voltage and temperature. In order to compare the changes in various parameters with temperature, voltage and current variations, the h -parameters are normalized to 1. Figure 3.20 shows the variation in h -parameters (in CE mode) with collector current for $I_C = 1$ mA, $V_{CE} = 5$ V, $T = 25^\circ\text{C}$ and $f = 1$ kHz.

The axis are graduated on a logarithmic scale to accommodate wide variations. The operating point is at $V_{CE} = 5$ V and $I_C = 1$ mA and hence all the parametric curves intersect at this point. Observe from Fig. 3.20 that at $I_C = 0.1$ mA, h_{fe} is 50% of its value at $I_C = 1$ mA. Thus, if h_{fe} is 100 at $I_C = 1$ mA, it would be 50 at $I_C = 0.1$ mA. At $I_C = 3$ mA, h_{fe} is 150% of its value at $I_C = 1$ mA. If h_{fe} is 100 at $I_C = 1$ mA, it would be 150 at $I_C = 3$ mA.

Further, observe the variation in h_{re} . The value of h_{re} increases to nearly 10 times its value at $I_C = 1$ mA, while h_{oe} increases to nearly 40 times its value at $I_C = 1$ mA. At such values h_{oe} may become large enough in comparison with load resistance to be ignored while approximating the model. Figure 3.21 shows the variation of h -parameters with collector-emitter voltage.

The variations are plotted with reference to the same operating point as in the previous figure. Observe that h_{fe} and h_{ie} do not change much in comparison with h_{re} and h_{oe} . Note that values of h_{oe} and h_{re} increase significantly on either sides of the operating point. Further, both with respect to

changes in collector current as well as collector voltage, variation in h_{fe} is minimal when compared to other parameters.

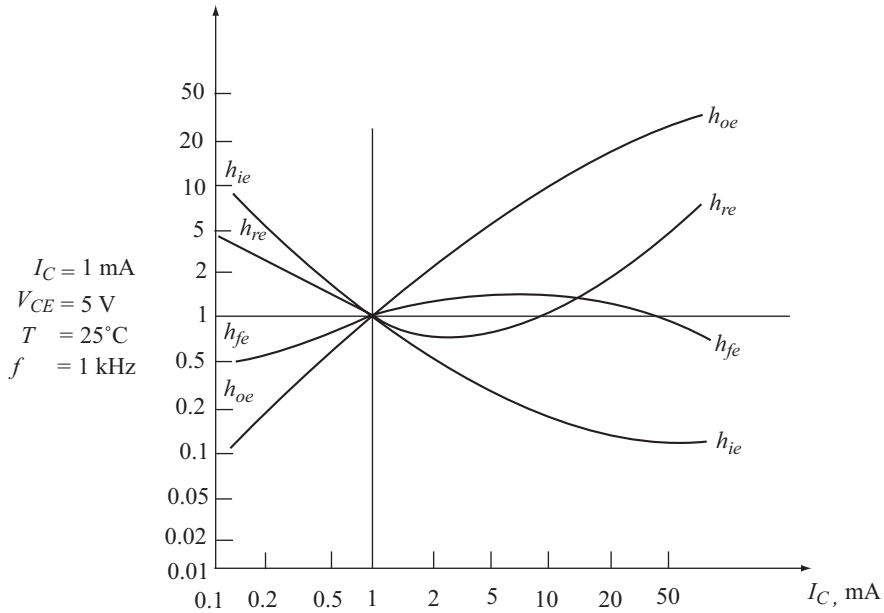


Fig. 3.20 Variation of CE h-parameters with collector current

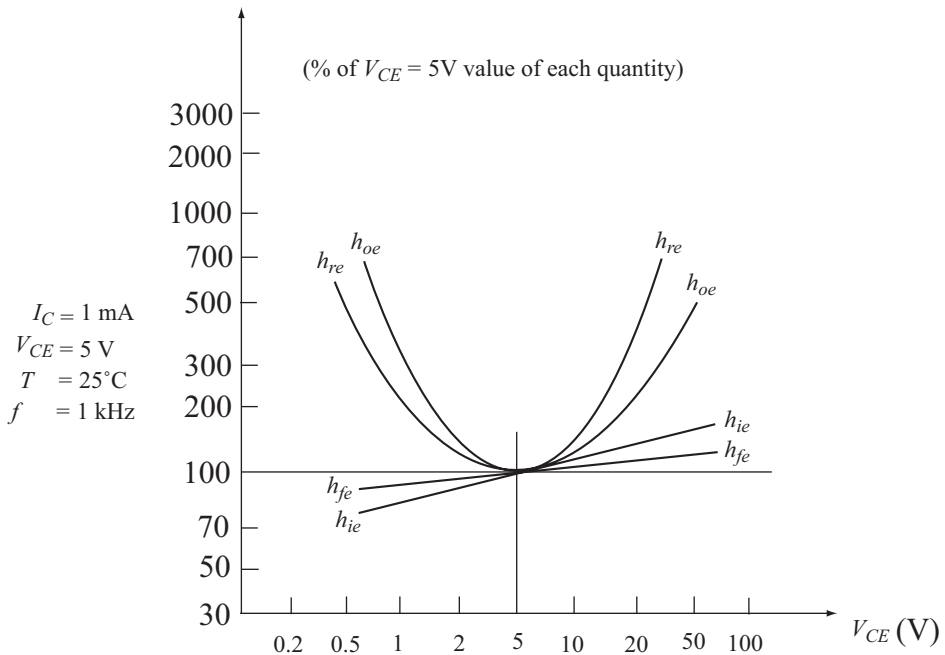


Fig. 3.21 Variation of CE h-parameters with collector-emitter voltage

Figure 3.22 shows the variation of h -parameters with respect to temperature. Observe that h_{oe} is least affected by the changes in temperature. However, in general, the values of all parameters increase with temperature. The value of input impedance h_{ie} varies the most with temperature. The value of h_{fe} drops to half its value at 25°C at a temperature of -50°C and, at 150°C it increases to 150% of its value at 25°C . All these variations must be taken into consideration when designing circuits using transistors.

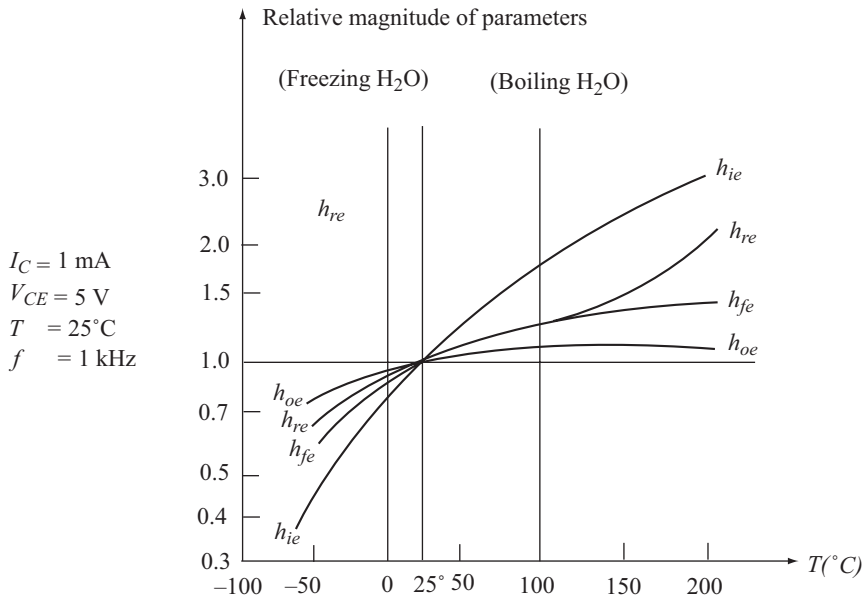


Fig. 3.22 Variation of CE h -parameters with temperature

◆ 3.14 ADVANTAGES OF h -PARAMETERS

The advantages of h -parameters are :

- The h -parameters are extremely useful in the analysis and design of circuits using transistors.
- At audio frequencies the h -parameters are real numbers which makes computations easy. This is essentially true because all capacitances have been treated as open circuits at low-frequencies.
- These parameters can be easily determined from the transistor characteristics.
- Since, we started the definition of h -parameters, by considering the transistor as a general two-port, four-terminal device, the models developed are independent of whether the transistors are of nnp or pnp type.
- Further, once the h -parameters are graphically obtained for one configuration, the h -parameters of other configurations can be obtained by a simple conversion.

◆ 3.15 APPROXIMATE HYBRID EQUIVALENT CIRCUIT

Consider the hybrid equivalent circuit of Fig. 3.13. For the CE and CB configurations, the magnitude of h_r and h_o is such that the results obtained for the important quantities such as Z_o , Z_i , A_v and A_i are only slightly affected if h_r and h_o are not included in the model.

From Table 3.5 we find that

$$h_{re} = 2.5 \times 10^{-4} \quad \text{and} \quad h_{rb} = 3.0 \times 10^{-4}$$

Due to very small values of h_{re} and h_{rb} we can take h_{re} and h_{rb} approximately equal to zero. As a result $h_r V_o = 0$. Thus the controlled voltage source $h_r V_o$ can be replaced by a short circuit as shown in Fig. 3.23.

Also from Table 3.5 we find that

$$1 / h_{oe} = 40 \text{ k}\Omega \quad \text{and} \quad 1 / h_{ob} = 2 \text{ M}\Omega$$

In practical situations, these values are quite large when compared with the parallel load R_L which will be connected between the output terminals. Thus $1 / h_o$ can be replaced by an open circuit as shown in Fig. 3.23.

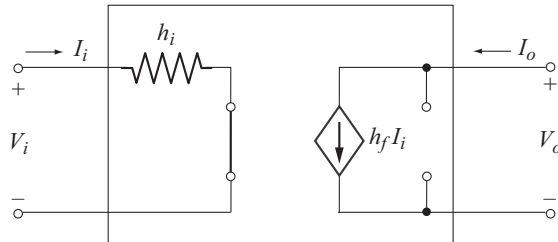


Fig. 3.23 Removing h_r and h_o from the hybrid equivalent model

Figure 3.24 shows the approximate hybrid equivalent circuit.

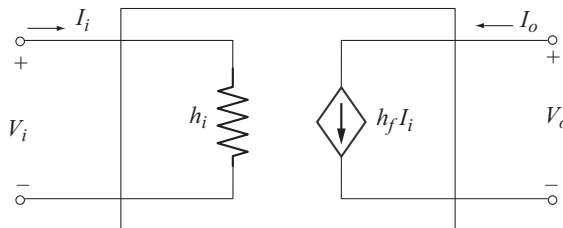


Fig. 3.24 Approximate hybrid equivalent circuit

The approximate hybrid equivalent circuit for CE configuration is shown in Fig. 3.25.

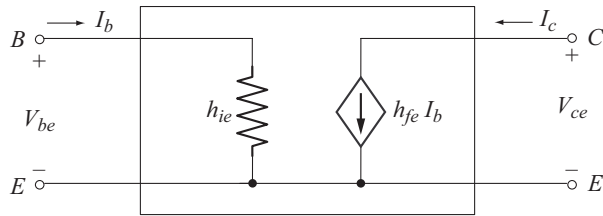


Fig. 3.25 Approximate hybrid equivalent circuit for CE configuration

Figure 3.26 shows the approximate hybrid equivalent circuit for CB configuration.

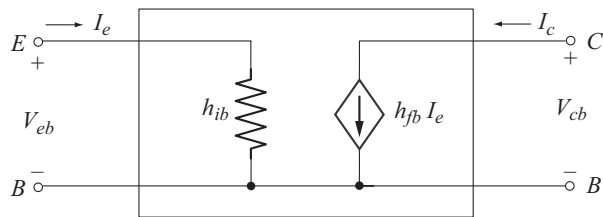


Fig. 3.26 Approximate hybrid equivalent circuit for CB configuration

◆ 3.16 RELATION BETWEEN THE PARAMETERS OF HYBRID MODEL AND THE r_e MODEL

Figure 3.27 shows the comparison between the approximate hybrid model and the r_e model of the transistor in CE configuration.

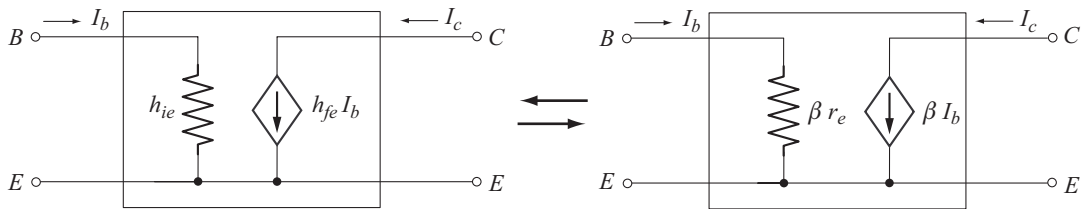


Fig. 3.27 Hybrid model versus r_e model for CE configuration

Comparing the two models we obtain the following relations:

$$h_{ie} = \beta r_e \quad (3.35)$$

$$\text{and} \quad h_{fe} = \beta \quad (3.36)$$

In Fig. 3.28 the approximate hybrid model and r_e model of the transistor in CB configuration are compared.

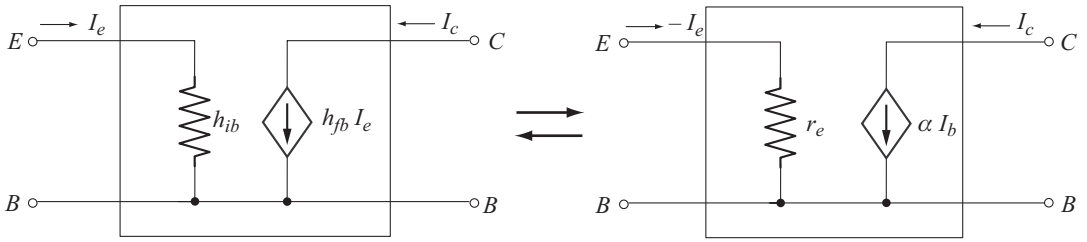


Fig. 3.28 Hybrid model versus r_e model for CB configuration

Comparing the two models we obtain the following relations:

$$h_{ib} = r_e \quad (3.37)$$

$$h_{fb} = \alpha$$

In any transistor (either pnp or npn) I_e and I_c are in opposite directions. Therefore h_{fb} , which is the ratio of I_c to I_e must have negative sign. But the value of α is given with positive sign. Hence it is appropriate to write.

$$h_{fb} = -\alpha \quad (3.38)$$

Since $\alpha \approx 1$, we can write

$$h_{fb} = -\alpha \approx -1 \quad (3.39)$$

3.16.1 Conversion between Hybrid Parameters

Having obtained the hybrid parameters for one transistor configuration, it is easy to find the parameters for other transistor configurations by a simple conversion.

Usually the CB and CC parameters are expressed in terms of the popular CE parameters as given below.

CB Hybrid Parameters in Terms of CE Hybrid Parameters

$$h_{ib} = \frac{h_{ie}}{1 + h_{fe}} \quad (3.40)$$

$$h_{rb} = \frac{h_{ie} h_{oe}}{1 + h_{fe}} - h_{re} \quad (3.41)$$

$$h_{fb} = \frac{-h_{fe}}{1 + h_{fe}} \quad (3.42)$$

$$h_{ob} = \frac{h_{oe}}{1 + h_{fe}} \quad (3.43)$$

CC Hybrid Parameters in Terms of CE Hybrid Parameters

$$h_{ic} = h_{ie} \quad (3.44)$$

$$h_{rc} = 1 - h_{re} \quad (3.45)$$

$$h_{fc} = -(1 + h_{fe}) \quad (3.46)$$

$$h_{oc} = h_{oe} \quad (3.47)$$

Example 3.2

Given $I_E = 3.2 \text{ mA}$, $h_{fe} = 150$, $r_o = \frac{1}{h_{oe}} = 40 \text{ k}\Omega$ and $h_{ob} = 0.5 \text{ }\mu\text{S}$, determine

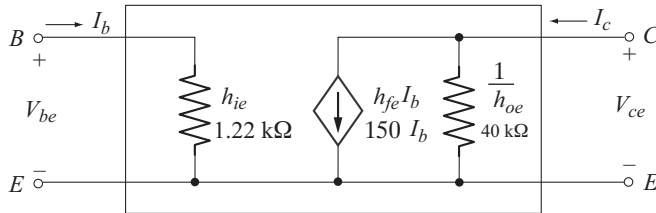
- The common-emitter hybrid equivalent circuit
- The common-base r_e model.

Solution

$$(a) \quad r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{3.2 \text{ mA}} = 8.125 \text{ }\Omega$$

$$h_{ie} = \beta r_e = (150)(8.125 \text{ }\Omega) = 1.22 \text{ k}\Omega$$

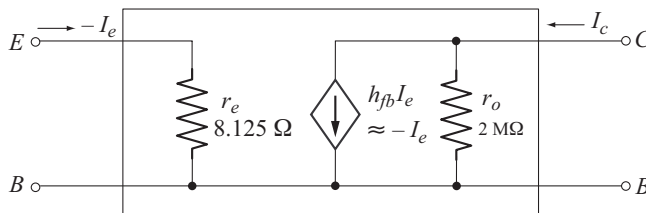
The common-emitter hybrid equivalent circuit is shown below.



$$(b) \quad r_o = \frac{1}{h_{ob}} = \frac{1}{0.5 \text{ }\mu\text{S}} = 2 \text{ M}\Omega$$

$$h_{fb} = \frac{-h_{fe}}{1 + h_{fe}} = \frac{-150}{151} = -0.993 \approx -1$$

The common-base r_e model is shown below.



◆ 3.17 HYBRID π MODEL

At high frequencies the effect of junction capacitances of the transistor has to be considered since they conduct appreciable amount of current due to their low reactance. The high frequency model of the transistor in CE configuration in which these capacitive effects are taken into consideration is called the hybrid π model or GIACOLETTO MODEL and is shown in Fig. 3.29.

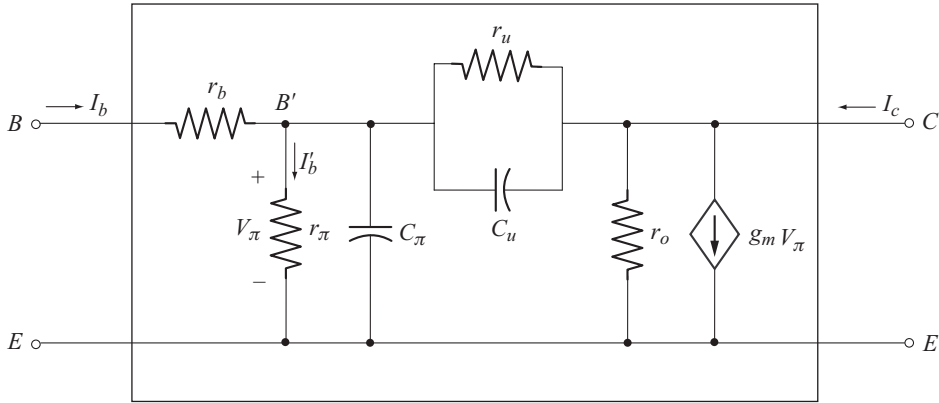


Fig. 3.29 The hybrid- π model of transistor in CE configuration

Let us discuss the significance of each component in the model shown in Fig. 3.29.

Hybrid π Capacitances

In chapter, 1, we have seen that a forward biased p-n junction exhibits diffusion capacitance and a reverse biased p-n junction exhibits transition capacitance.

In the model of Fig. 3.29, C_π represents the diffusion capacitance of forward-biased emitter-base junction. This capacitance is an indicative of excess minority carrier storage in the base. C_π is usually just a few picofarads to a few tens of picofarads.

C_u represents the transition capacitance of the reverse biased collector-base junction. C_u typically extends from less than 1 pF to a few Picofarads.

C_π and C_u are alternatively represented by C_{be} and C_{bc} respectively.

Hybrid π Resistances

The internal node B' is not physically accessible as shown in Fig. 3.30. Node B represents the external base terminal. The resistance r_b includes the following.

- the base contact resistance
- the base bulk resistance and
- the base spreading resistance.

It is typically a few ohms to tens of ohms.

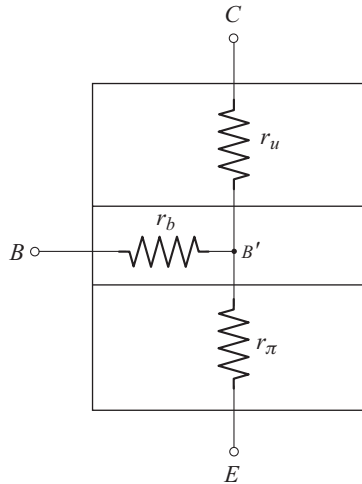


Fig. 3.30 Hybrid π resistances

The resistors r_π , r_u and r_o are the resistances between the indicated terminals of the transistor when the transistor is in the active region.

The resistance r_π is identical to βr_e introduced in the common-emitter r_e model. The resistance r_u represents the feedback from collector to base. It is very large typically in the mega ohm range.

r_o is the output resistance which appears between the collector and emitter terminals. Typically it lies in the range of 5 k Ω to 40 k Ω and it is determined from the hybrid parameter h_{oe} .

Hybrid π Conductance

The small signal collector current, due to small change in the voltage V_π , with the collector shorted to emitter is accounted by the voltage controlled current source $g_m V_\pi$ connected between the collector and emitter terminals.

The subscripts π and u associated with hybrid π parameters have the following meaning.

π : comes from the hybrid π terminology.

u : the element with this subscript provides union between collector and base terminals.

Hybrid π Parameters in Terms of r_e and h Parameters

The following relations can be used to find the parameters of hybrid π model using r_e and the h parameters.

$$r_\pi = \beta r_e \quad (3.48)$$

$$g_m = \frac{1}{r_e} \quad (3.49)$$

$$r_o = \frac{1}{h_{oe}} \quad (3.50)$$

$$h_{re} = \frac{r_{\pi}}{r_{\pi} + r_u} \approx \frac{r_{\pi}}{r_u} \quad (3.51)$$

3.17.1 Low Frequency Hybrid Model from Hybrid π Model

For low-to-mid frequency analysis, the effect of capacitances C_{π} and C_u can be neglected due to the very high reactance associated with each of them.

The resistance r_b is usually so small, it can be replaced by a short-circuit. The resistance r_u usually so large, it can be treated as an open circuit.

The simplified low frequency hybrid π model is shown in Fig. 3.31.

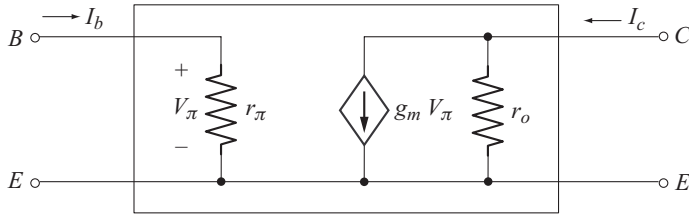


Fig. 3.31 Simplified low frequency hybrid π model

$$\begin{aligned} r_{\pi} &= \beta r_e = h_{ie} \\ r_o &= 1 / h_{oe} \\ V_{\pi} &= I_b r_{\pi} = I_b \beta r_e \\ g_m V_{\pi} &= \frac{1}{r_e} (I_b \beta r_e) = \beta I_b = h_{fe} I_b \quad (\because \beta = h_{fe}) \end{aligned}$$

The resulting model is shown in Fig. 3.32 which is same as the low frequency hybrid model with h_{re} taken equal to zero.

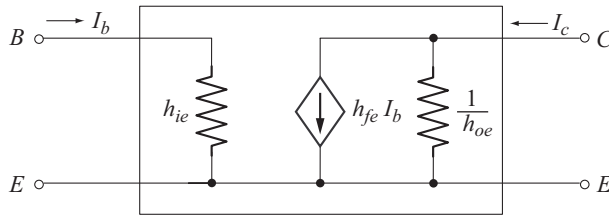


Fig. 3.32 Low frequency hybrid model with h_{re} removed

Example 3.3

(a) Sketch the Giacoleto (hybrid π) model for a common-emitter transistor given that:

$$\begin{array}{lll} r_b = 4 \, \Omega & C_{\pi} = 5 \, \text{pF} & C_u = 1.5 \, \text{pF} \\ h_{oe} = 18 \, \mu\text{S} & \beta = 120 & r_e = 14 \, \Omega \end{array}$$

- (b) If the applied load is $1.2 \text{ k}\Omega$ and the source resistance is $250 \text{ }\Omega$, draw the approximate hybrid π model for low and mid frequency range.

Solution

$$(a) \quad r_{\pi} = \beta r_e = (120)(14 \text{ }\Omega) = 1.68 \text{ k}\Omega$$

$$g_m = \frac{1}{r_e} = \frac{1}{14 \text{ }\Omega} = 0.0714 \text{ S}$$

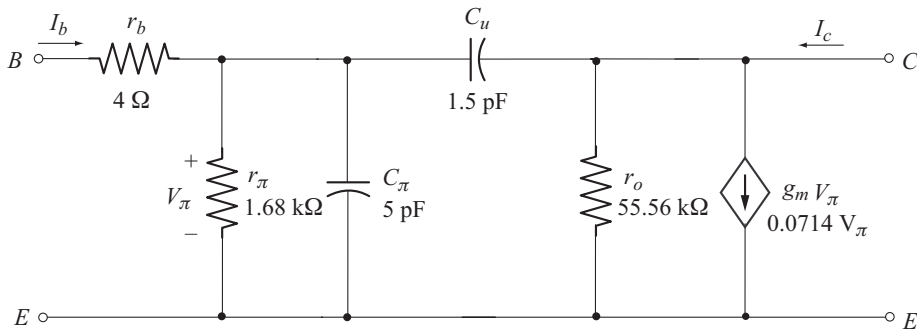
$$r_o = \frac{1}{h_{oe}} = \frac{1}{18 \times 10^{-6}} = 55.56 \text{ k}\Omega$$

$$h_{re} \approx \frac{r_{\pi}}{r_u} \Rightarrow r_u = \frac{r_{\pi}}{h_{re}}$$

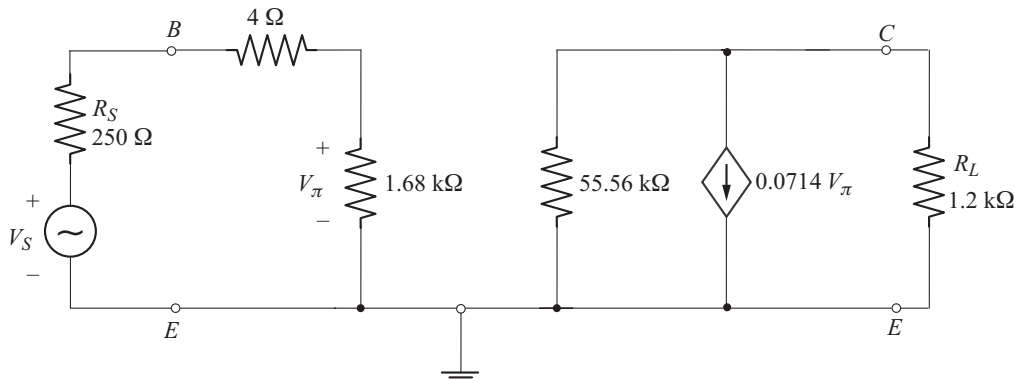
Since h_{re} is not given, let us take $h_{re} = 0$

$$\therefore r_u = \infty \Rightarrow \text{open circuit.}$$

The hybrid π model is shown below.



- (b) For low and mid frequency range of operation, the capacitors C_u and C_{π} can be replaced by open circuit equivalent due their high reactance. The simplified hybrid π model is shown below.



◆ 3.18 COMMON-EMITTER FIXED-BIAS CONFIGURATION

Figure 3.33 shows the common-emitter amplifier using fixed bias. The ac input signal V_i is applied to the base of transistor through the input coupling capacitor C_1 . The amplified signal V_o is taken at the collector through the output coupling capacitor C_2 .

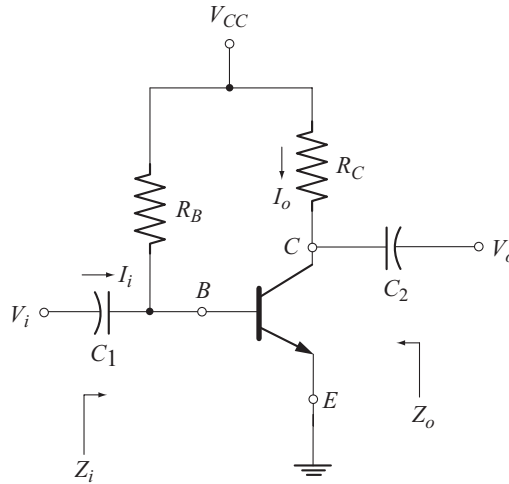


Fig. 3.33 Common-emitter fixed bias configuration

In order to perform the small-signal ac analysis let us obtain the ac equivalent circuit by reducing the dc source V_{CC} to zero and short circuiting the coupling capacitors C_1 and C_2 as shown in Fig. 3.34.

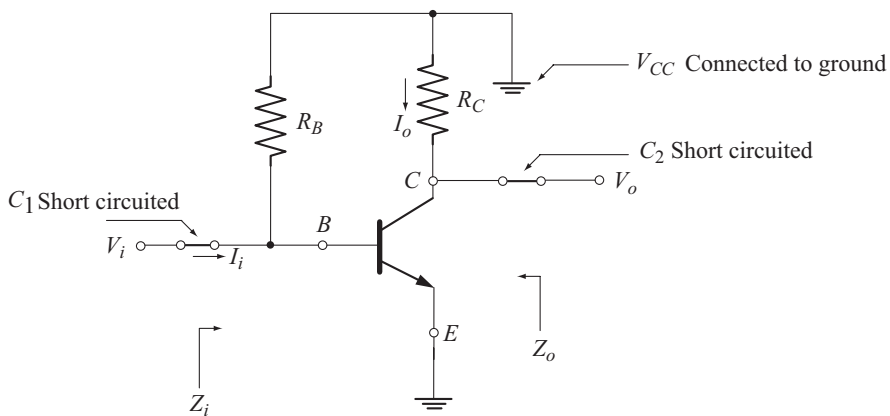


Fig. 3.34 AC equivalent circuit

Note that R_B appears between base and ground and R_C between collector and ground. The ac equivalent circuit is redrawn in Fig. 3.35.

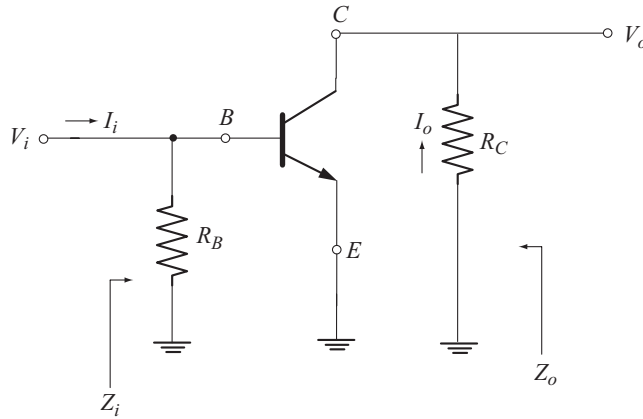


Fig. 3.35 AC equivalent circuit redrawn

Note that the emitter terminal is common to input and output circuits. Hence the configuration is common-emitter. Let us replace the transistor by its common emitter r_e model as shown in Fig. 3.36.

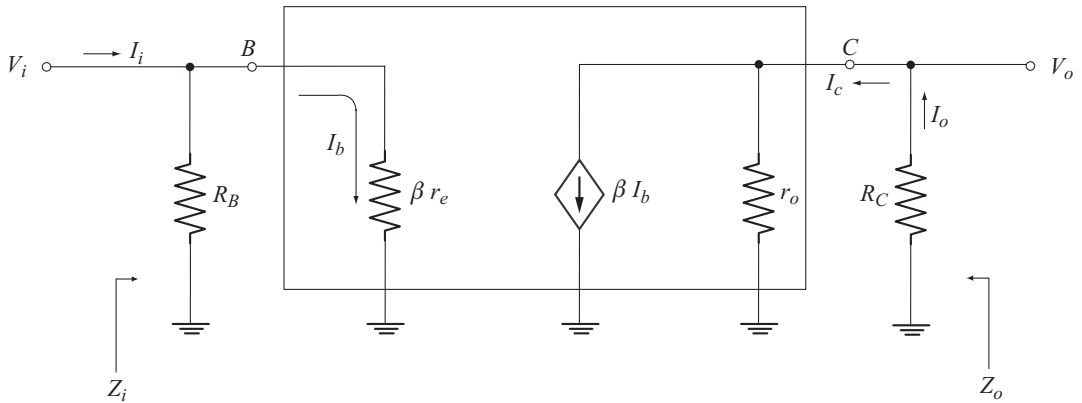


Fig. 3.36 AC equivalent circuit with r_e model

3.18.1 Input Impedance (Z_i)

The input impedance is given by

$$Z_i = \frac{V_i}{I_i} \quad (3.52)$$

From the input circuit of Fig. 3.36, we find that, Z_i is given by the parallel combination of R_B and βr_e .

$$\therefore Z_i = R_B \parallel \beta r_e \quad (3.53)$$

3.18.2 Output Impedance (Z_o)

To find the output impedance we have to reduce V_i to zero.

With $V_i = 0$, $I_i = 0$, $I_b = 0$ and therefore $\beta I_b = 0$. Thus we have to open circuit the current source. The output equivalent circuit under this condition is shown in Fig. 3.37.

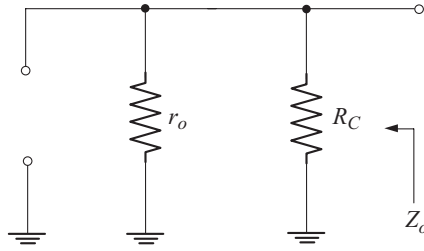


Fig. 3.37 Circuit to find Z_o

Note that Z_o is given by the parallel combination of r_o and R_C

$$\therefore Z_o = r_o \parallel R_C \quad (3.54)$$

3.18.3 Voltage Gain (A_v)

To find the voltage gain let us consider the output equivalent circuit shown in Fig. 3.38.

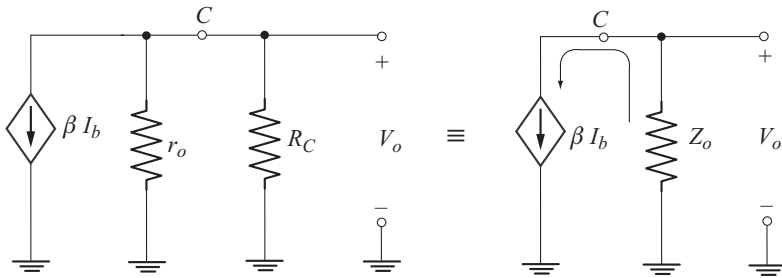


Fig. 3.38 Circuit to find V_o

$$\begin{aligned} V_o &= -\beta I_b Z_o \\ &= -\beta I_b [r_o \parallel R_C] \end{aligned} \quad (3.55)$$

From the input circuit of Fig. 3.36 we have

$$V_i = I_b [\beta r_e] \quad (3.56)$$

Now the voltage gain is

$$A_v = \frac{V_o}{V_i}$$

$$= \frac{-\beta I_b [r_o \parallel R_C]}{I_b [\beta r_e]}$$

$$A_V = -\frac{r_o \parallel R_C}{r_e} \quad (3.57)$$

If $R_B \geq 10 \beta r_e$, $R_B \parallel \beta r_e \approx \beta r_e$

Now Equation (3.53) become

$$Z_i \approx \beta r_e \quad (3.58)$$

For $r_o \geq 10 R_C$

Under this situation

$$R_C \parallel r_o \approx R_C$$

From Equation (3.54)

$$Z_o \approx R_C \quad (3.59)$$

From Equation (3.57)

$$A_V \approx -\frac{R_C}{r_e} \quad (3.60)$$

3.18.4 Current Gain (A_I)

The current gain is defined by

$$A_I = \frac{I_o}{I_i} \quad (3.61)$$

$$V_o = -I_o R_C \quad (\text{From Fig. 3.36})$$

$$\therefore I_o = \frac{-V_o}{R_C} \quad (3.62)$$

$$\text{Also } I_i = \frac{V_i}{Z_i} \quad (3.63)$$

Using these relations in Equation (3.61) we have

$$A_V = \left[\frac{-V_o}{R_C} \right] \div \left[\frac{V_i}{Z_i} \right]$$

$$= - \left[\frac{V_o}{V_i} \right] \left[\frac{Z_i}{R_C} \right]$$

$$A_I = -\frac{A_V Z_i}{R_C} \quad (3.64)$$

Using Equations (3.60) and (3.58) for A_v and Z_i respectively, we obtain

$$\begin{aligned} A_I &\approx - \left[\frac{1}{R_C} \right] \left(\frac{-R_C}{r_e} \right) (\beta r_e) \\ A_I &\approx \beta \end{aligned} \quad (3.65)$$

3.18.5 Phase Relationship

The negative sign in the equation for A_v implies that the input signal V_i and the output signal V_o are 180° out of phase as shown in Fig. 3.39.

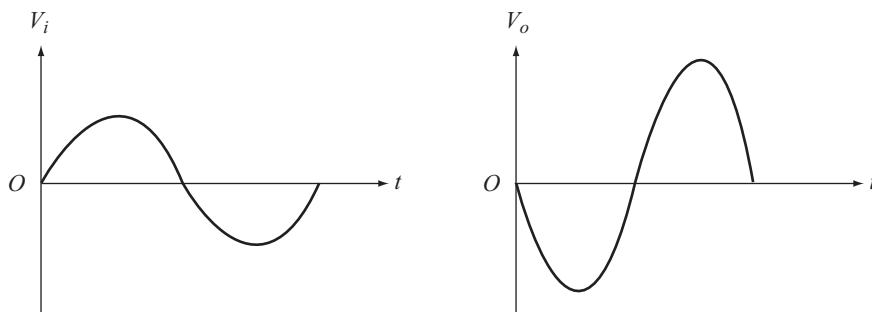
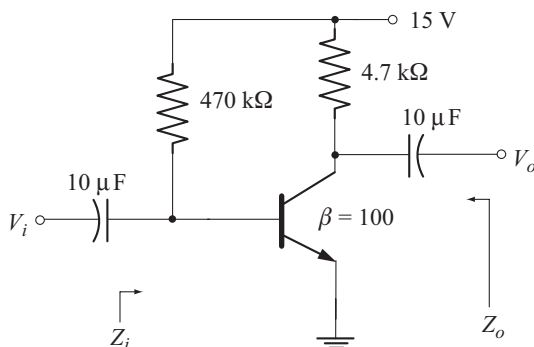


Fig. 3.39 180° Phase shift between V_i and V_o

Example 3.4

For the circuit shown below:

- Determine r_e
- Find Z_i , Z_o , A_v and A_I (with $r_o = \infty \Omega$)
- Repeat part (b) taking $r_o = 50 \text{ k}\Omega$ and compare the results.



Solution

$$\begin{aligned} \text{(a)} \quad r_e &= \frac{26 \text{ mV}}{I_E} \\ I_E &= (1 + \beta) I_B \end{aligned}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{15 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega} = 30.43 \text{ }\mu\text{A}$$

$$I_E = (101)(30.43 \text{ }\mu\text{A}) = 3.073 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{3.073 \text{ mA}} = 8.46 \text{ }\Omega$$

(b)

$$Z_i = R_B \parallel \beta r_e$$

$$\beta r_e = (100)(8.46 \text{ }\Omega) = 846 \text{ }\Omega$$

$$Z_i = 470 \text{ k}\Omega \parallel 846 \text{ }\Omega = 844.47 \text{ }\Omega$$

$$Z_o = r_o \parallel R_C = \infty \text{ }\Omega \parallel 4.7 \text{ k}\Omega = 4.7 \text{ k}\Omega$$

$$A_V = -\frac{r_o \parallel R_C}{r_e} = -\frac{4.7 \text{ k}\Omega}{8.46 \text{ }\Omega} = -555.55$$

$$A_I = \frac{-A_V Z_i}{R_C} = \frac{-(-555.55)(844.47 \text{ }\Omega)}{4.7 \text{ k}\Omega} = 99.82$$

(c) When

 Z_i is not affected by r_o $\therefore$

$$Z_i = 844.47 \text{ }\Omega. \quad (\text{As calculated in part (b)})$$

$$Z_o = 50 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega = 4.3 \text{ k}\Omega$$

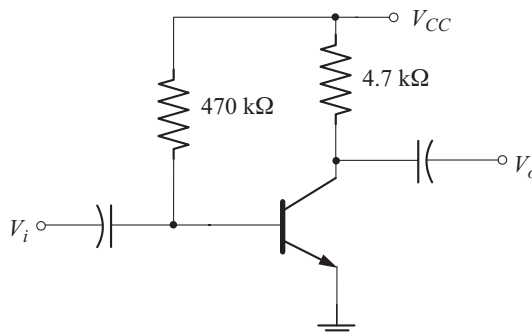
$$A_V = \frac{-4.3 \text{ k}\Omega}{8.46 \text{ }\Omega} = -508.27$$

$$A_I = -\frac{(-508.27)(844.47 \text{ }\Omega)}{(4.7 \text{ k}\Omega)} = 91.32$$

Note that the values of Z_o and A_V with finite r_o are less than that with $r_o = \infty$.

Example 3.5

For the amplifier circuit shown below calculate V_{CC} for a voltage gain of $A_V = -200$. Take $\beta = 90$ and $r_o = \infty \text{ }\Omega$.



Solution

$$A_V = - \frac{r_o \parallel R_C}{r_e}$$

$$R_C \parallel r_o = 4.7 \text{ k}\Omega \parallel \infty \Omega \\ = 4.7 \text{ k}\Omega$$

$$-200 = \frac{-4.7 \text{ k}\Omega}{r_e}$$

$$r_e = 23.5 \Omega$$

$$r_e = \frac{26 \text{ mV}}{I_E}$$

$$I_E = \frac{26 \text{ mV}}{23.5 \Omega} = 1.106 \text{ mA}$$

$$I_E = (1 + \beta) I_B$$

$$I_B = \frac{I_E}{1 + \beta} = \frac{1.106 \text{ mA}}{91} = 12.15 \mu\text{A}$$

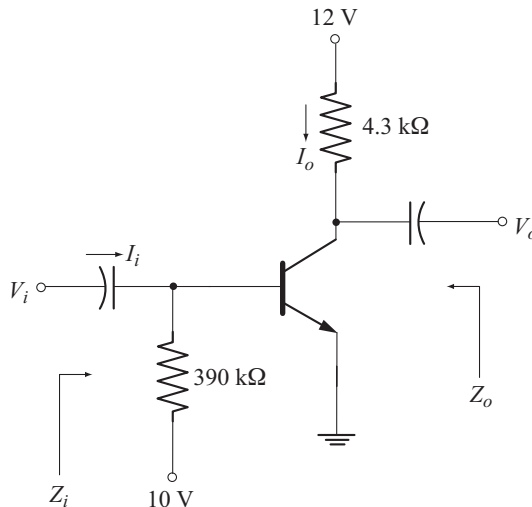
$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$\Rightarrow V_{CC} = I_B R_B + V_{BE} = (12.15 \mu\text{A})(470 \text{ k}\Omega) + 0.7 \text{ V} = 6.4 \text{ V}$$

Example 3.6

For the circuit shown below, taking $\beta = 100$ and $r_o = 60 \text{ k}\Omega$, calculate

- (a) r_e
- (b) Z_i and Z_o
- (c) A_V
- (d) the effect of $r_o = 30 \text{ k}\Omega$ on A_V



- (a)
$$I_B = \frac{10\text{V} - V_{BE}}{390\text{k}\Omega} = \frac{10\text{V} - 0.7\text{V}}{390\text{k}\Omega} = 23.85\text{ }\mu\text{A}$$

$$I_E = (1 + \beta) I_B = (101) (23.85\text{ }\mu\text{A}) = 2.4\text{ mA}$$

$$r_e = \frac{26\text{mV}}{I_E} = \frac{26\text{mV}}{2.4\text{mA}} = 10.83\text{ }\Omega$$
- (b) From the given figure, $R_B = 390\text{ k}\Omega$ and $R_C = 4.3\text{ k}\Omega$

$$Z_i = \beta r_e = (100)(10.83\text{ }\Omega) = 1.083\text{ k}\Omega$$

$$Z_o = R_C \parallel r_o = 4.3\text{ k}\Omega \parallel 60\text{ k}\Omega = 4.012\text{ k}\Omega$$
- (c)
$$A_V = - \frac{R_C \parallel r_o}{r_e} = - \frac{4.012\text{ k}\Omega}{10.83\text{ }\Omega} = -370.45$$
- (e) when $r_o = 30\text{ k}\Omega$

$$R_C \parallel r_o = 4.3\text{ k}\Omega \parallel 30\text{ k}\Omega = 3.76\text{ k}\Omega$$

$$A_V = - \frac{3.76\text{ k}\Omega}{10.83\text{ }\Omega} = -347.18$$

◆ 3.19 COMMON EMITTER CONFIGURATION WITH VOLTAGE DIVIDER BIAS

Figure 3.40 shows CE configuration using voltage divider bias.

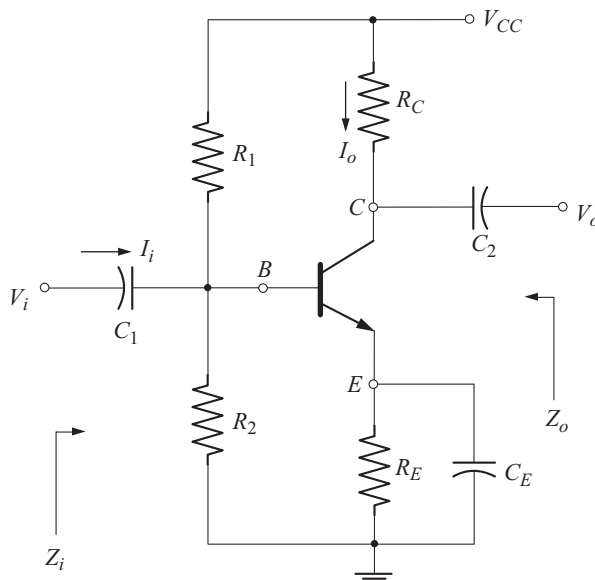


Fig. 3.40 CE configuration using voltage divider bias

The ac input signal V_i is applied to the base of the transistor through the input coupling capacitor C_1 . The amplified signal V_o is taken at the collector through the output coupling capacitor C_2 . The emitter bypass capacitor C_E is used to prevent the loss of voltage gain due to ac negative feedback through R_E by creating an ac ground at the emitter. C_1 , C_2 and C_E are so selected that they represent short circuit even at the lowest frequency of operation.

To perform small signal ac analysis, let us obtain the ac equivalent circuit by reducing V_{CC} to zero and replacing the capacitors C_1 , C_2 and C_E by short circuits as shown in Fig. 3.41.

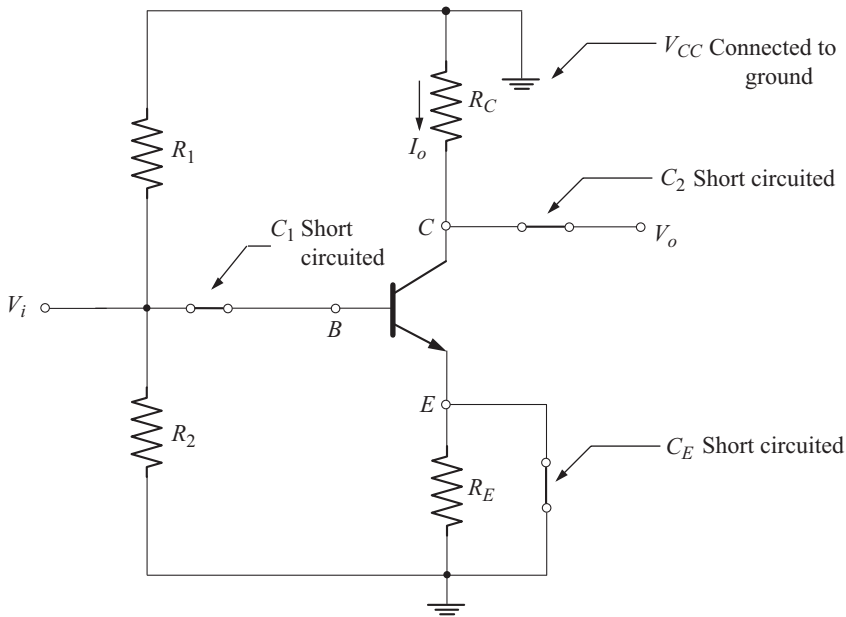


Fig. 3.41 AC equivalent circuit

The following observations can be made from Fig. 3.41.

- (a) R_1 appears between base and ground. Thus R_1 comes in parallel with R_2 .
- (b) R_C appears between collector and ground
- (c) R_E is in parallel with a short circuit (0Ω)

$$\therefore R_E \parallel 0 \Omega = 0 \Omega$$

i.e., the equivalent is a short circuit.

The ac equivalent circuit is redrawn in Fig. 3.42.

Note that the emitter terminal is common to input and output circuits. Hence the configuration is common-emitter.

$$\text{Let } R' = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

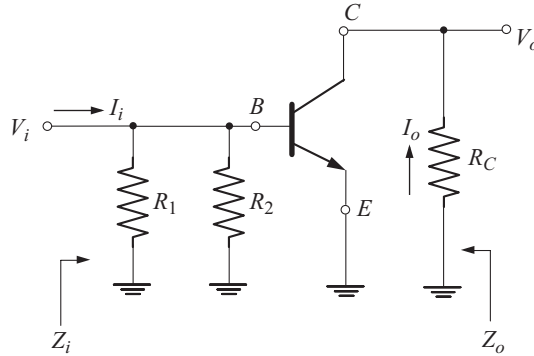


Fig. 3.42 AC equivalent circuit redrawn

The ac equivalent circuit of Fig. 3.42 is an exact duplicate of the equivalent circuit shown in Fig. 3.35 for fixed bias configuration. The only exception is that, R_B is replaced by R' . Therefore, the results derived in the previous section can be readily applied to the circuit under consideration. The results are reproduced here for convenience.

- $$Z_i = \frac{V_i}{I_i} = R' \parallel \beta r_e \quad (3.66)$$

- $$Z_o = r_o \parallel R_C \quad (3.67)$$

- $$A_V = -\frac{r_o \parallel R_C}{r_e} \quad (3.68)$$

- $$A_I = -\frac{A_V Z_i}{R_C} \quad (3.69)$$

For $r_o \geq 10 R_C$

- $$Z_o \approx R_C \quad (3.70)$$

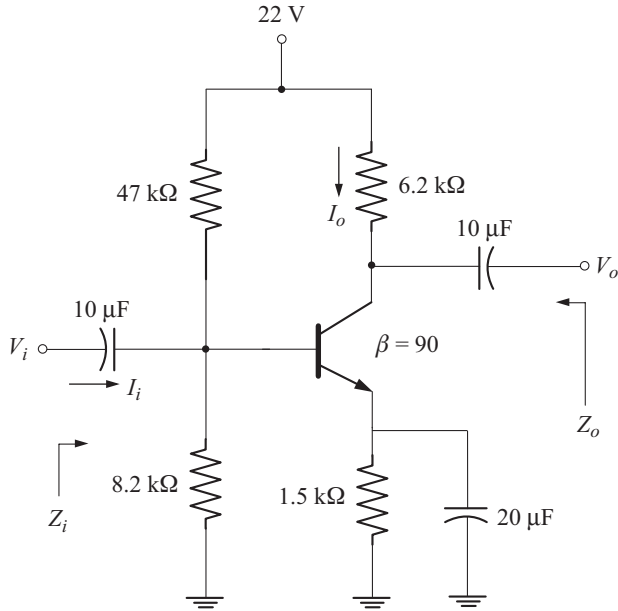
- $$A_V \approx -\frac{R_C}{r_e} \quad (3.71)$$

Negative sign in A_V reveals that, V_i and V_o are 180° out of phase.

Example 3.7

For the circuit shown below, taking $r_o = \infty \Omega$, calculate:

- (a) r_e
- (b) Z_i
- (c) Z_o
- (d) A_V and A_I
- (e) Repeat part (b) to (d) with $r_o = \frac{1}{h_{oe}} = 50 \text{ k}\Omega$ and compare the results.



Check for $\beta R_E \geq 10 R_2$

$$\beta R_E = (90)(1.5 \text{ k}\Omega) = 135 \text{ k}\Omega$$

$$10 R_2 = (10)(8.2 \text{ k}\Omega) = 82 \text{ k}\Omega$$

Since $\beta R_E > 10 R_2$, we can use approximate analysis

$$R_{Th} = R' = R_1 \parallel R_2 = 47 \text{ k}\Omega \parallel 8.2 \text{ k}\Omega = 6.98 \text{ k}\Omega$$

$$V_B = V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{(22 \text{ V})(8.2 \text{ k}\Omega)}{47 \text{ k}\Omega + 8.2 \text{ k}\Omega} = 3.268 \text{ V}$$

$$V_{Th} = V_{BE} + V_E \Rightarrow V_E = V_{Th} - V_{BE}$$

$$V_E = 3.268 \text{ V} - 0.7 \text{ V} = 2.568 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{2.568 \text{ V}}{1.5 \text{ k}\Omega} = 1.712 \text{ mA}$$

$$(a) \quad r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.712 \text{ mA}} = 15.18 \Omega$$

$$(b) \quad Z_i = R' \parallel \beta r_e = (6.98 \text{ k}\Omega) \parallel (90)(15.18 \Omega) = 1.152 \text{ k}\Omega$$

$$(c) \quad Z_o = r_o \parallel R_C = (\infty \Omega) \parallel R_C = R_C = 6.2 \text{ k}\Omega$$

$$(d) \quad A_V = -\frac{r_o \parallel R_C}{r_e} = -\frac{R_C}{r_e} = -\frac{6.2 \text{ k}\Omega}{15.18 \Omega} = -408.43$$

$$A_I = - \frac{A_V Z_i}{R_C} = - \frac{(-408.43)(1.19 \text{ k}\Omega)}{(6.2 \text{ k}\Omega)} = 78.39$$

(e) When $r_o = 50 \text{ k}\Omega$

Check for $r_o \geq 10 R_C$

$$10 R_C = (10)(6.2 \text{ k}\Omega) = 62 \text{ k}\Omega$$

Since $r_o < 62 \text{ k}\Omega$, we have to use exact analysis.

$$Z_i = 1.152 \text{ k}\Omega \quad (\text{same as before})$$

$$Z_o = 50 \text{ k}\Omega \parallel 6.2 \text{ k}\Omega = 5.52 \text{ k}\Omega$$

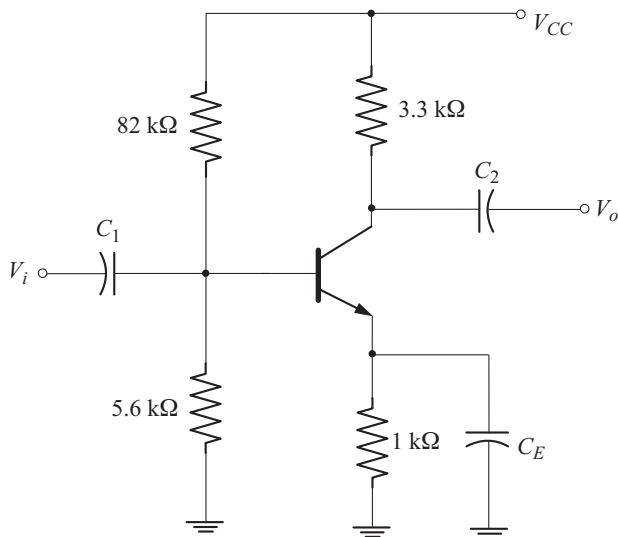
$$A_V = - \frac{5.52 \text{ k}\Omega}{15.18 \Omega} = -363.64$$

$$A_I = - \frac{(-363.64)(1.152 \text{ k}\Omega)}{(6.2 \text{ k}\Omega)} = 67.56$$

There is a considerable difference in the values of Z_o , A_V and A_I since the condition $r_o \geq 10 R_C$ is not satisfied.

Example 3.8

For the circuit shown below determine V_{CC} if $A_V = -160$ and $r_o = 100 \text{ k}\Omega$. Take $\beta = 100$



Solution

$$A_V = - \frac{r_o \parallel R_C}{r_e}$$

$$-160 = - \frac{100 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega}{r_e}$$

$$r_e = 19.96 \text{ }\Omega$$

$$r_e = \frac{26 \text{ mV}}{I_E} \Rightarrow I_E = \frac{26 \text{ mV}}{r_e}$$

$$I_E = \frac{26 \text{ mV}}{19.96 \text{ }\Omega} = 1.3 \text{ mA}$$

$$V_{Th} = V_{BE} + I_E R_E = 0.7 \text{ V} + (1.3 \text{ mA})(1 \text{ k}\Omega) = 2 \text{ V}$$

$$V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} \Rightarrow V_{CC} = \frac{V_{Th} (R_1 + R_2)}{R_2}$$

$$V_{CC} = \frac{(2 \text{ V})(82 \text{ k}\Omega + 5.6 \text{ k}\Omega)}{5.6 \text{ k}\Omega} = 31.2 \text{ V}$$

◆ 3.20 COMMON-EMITTER CONFIGURATION USING EMITTER-BIAS WITH UNBYPASSED R_E

Figure 3.43 shows CE configuration using emitter-bias. Note that the emitter resistor R_E is unbypassed.

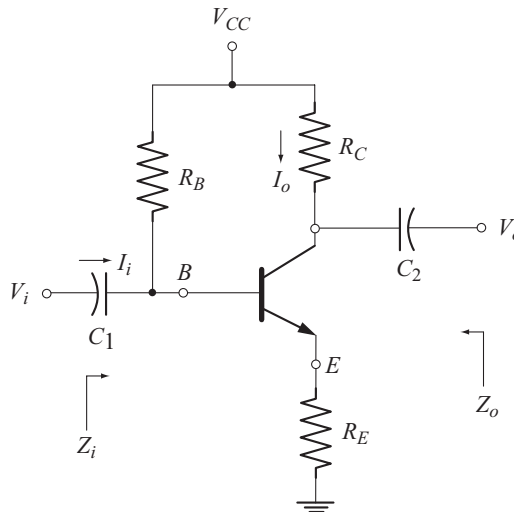


Fig. 3.43 CE configuration with emitter bias

The ac equivalent circuit is drawn by reducing V_{CC} to zero and replacing C_1 and C_2 by short circuit equivalent as shown in Fig. 3.44.

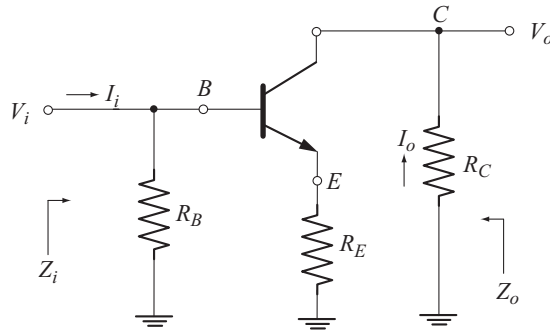


Fig. 3.44 AC equivalent circuit

Let us replace the transistor by its common-emitter r_e model as shown in Fig. 3.45. To simplify the analysis, r_o is not included in the model.

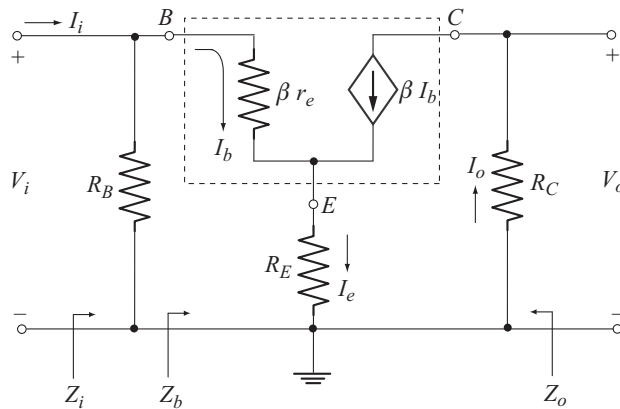


Fig. 3.45 AC equivalent circuit using r_e model

Input Impedance (Z_b)

Applying KVL to the input circuit of Fig. 3.45 we have

$$V_i = I_b \beta r_e + I_e R_E \quad (3.72)$$

using

$$I_e = (1 + \beta) I_b$$

$$V_i = I_b \beta r_e + (1 + \beta) I_b R_E$$

The input impedance looking into the network to the right of R_B (i.e., excluding R_B) is

$$Z_b = \frac{V_i}{I_b} = \beta r_e + (1 + \beta) R_E \quad (3.73)$$

Since $\beta \gg 1$,

$$(1 + \beta) \approx \beta$$

$$\begin{aligned} \therefore Z_b &\approx \beta r_e + \beta R_E \\ \text{or } Z_b &= \beta (r_e + R_E) \end{aligned} \quad (3.74)$$

Usually $r_e \ll R_E$.

Now Equation (3.74) can be further reduced to

$$Z_b \approx \beta R_E \quad (3.75)$$

Input Impedance (Z_i)

Z_i takes the effect of R_B into account. It is given by the parallel combination of R_B and Z_b as shown in Fig. 3.46.

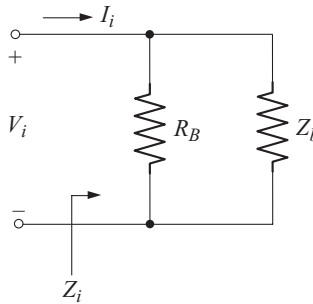


Fig. 3.46 Determining Z_i

$$Z_i = \frac{V_i}{I_i} = R_B \parallel Z_b \quad (3.76)$$

Output Impedance (Z_o)

To find Z_o , we have to set V_i to zero. With $V_i = 0$, $I_b = 0$ and therefore $\beta I_b = 0$. Thus the controlled current source can be replaced by an open circuit as shown in Fig. 3.47.

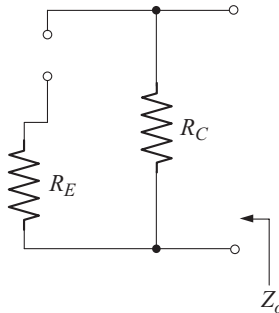


Fig. 3.47 Circuit to find Z_o

From Fig. 3.47 we find that Z_o is given by the parallel combination of R_C and ∞ ohms (open circuit).

$$\begin{aligned} Z_o &= R_C \parallel \infty \Omega \\ Z_o &= R_C \end{aligned} \quad (3.77)$$

Voltage Gain (A_V)

$$\begin{aligned} V_o &= -I_o R_C \\ &= -\beta I_b R_C \end{aligned}$$

From Equation (3.73),

$$\begin{aligned} I_b &= \frac{V_i}{Z_b} \\ \therefore V_o &= -\beta \left(\frac{V_i}{Z_b} \right) R_C \\ A_V &= \frac{V_o}{V_i} \\ &= -\frac{\beta R_C}{Z_b} \end{aligned} \quad (3.78)$$

Using, $Z_b = \beta(r_e + R_E)$ we have

$$A_V = -\frac{R_C}{r_e + R_E} \quad (3.79)$$

Taking, $r_e + R_E \approx r_e$ we get

$$A_V \approx -\frac{R_C}{R_E} \quad (3.80)$$

Note that A_V is β independent. As a result the voltage gain is independent of transistor, which is a desirable feature.

Current Gain (A_I)

$$\begin{aligned} I_o &= -\frac{V_o}{R_C} \\ \text{and} \quad I_i &= \frac{V_i}{Z_i} \\ \text{Now} \quad A_I &= \frac{I_o}{I_i} \\ &= -\left[\frac{V_o}{R_C} \right] \div \left[\frac{V_i}{Z_i} \right] \\ &= -\left[\frac{V_o}{V_i} \right] \left[\frac{Z_i}{R_C} \right] \end{aligned}$$

$$A_I = - \frac{A_V Z_i}{R_C} \quad (3.81)$$

Effect of Unbypassed R_E

Following are the effects of unbypassed R_E on the performance parameters of the amplifier.

- The input impedance increases by βR_E
- Voltage gain decreases
- Voltage gain is independent of β . Thus voltage gain is independent of the transistor and it depends only on the external components R_C and R_E . It means that voltage gain is stabilized.

The reason for these results is that, there is an ac negative feedback through R_E .

Phase Relationship

The negative sign in equation for A_V reveals that V_i and V_o are 180° out of phase.

◆ 3.21 CE EMITTER BIAS CONFIGURATION WITH BYPASSED R_E

Figure 3.48 shows CE emitter bias configuration with R_E bypassed by an emitter bypass capacitor C_E .

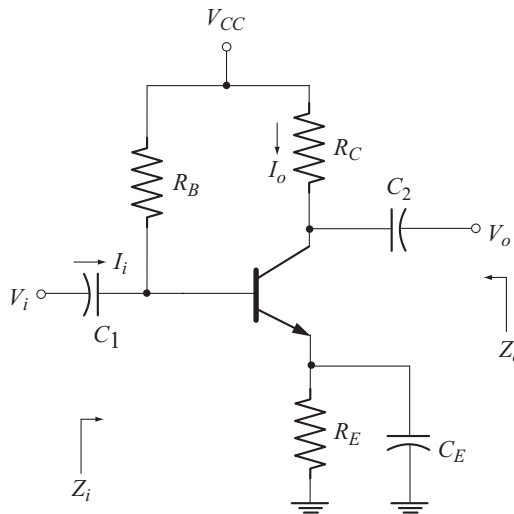


Fig. 3.48 CE emitter bias configuration with bypassed R_E

The ac equivalent circuit is shown in Fig. 3.49. Note that this ac equivalent circuit is an exact replica of that shown in Fig. 3.34, written for CE fixed bias configuration. Hence all the results derived in section 3.18 for CE fixed bias configuration can be readily applied to this circuit.

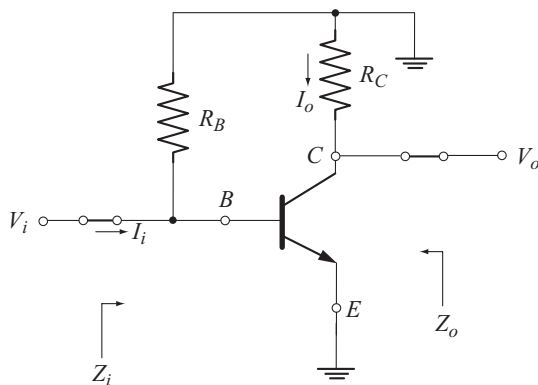


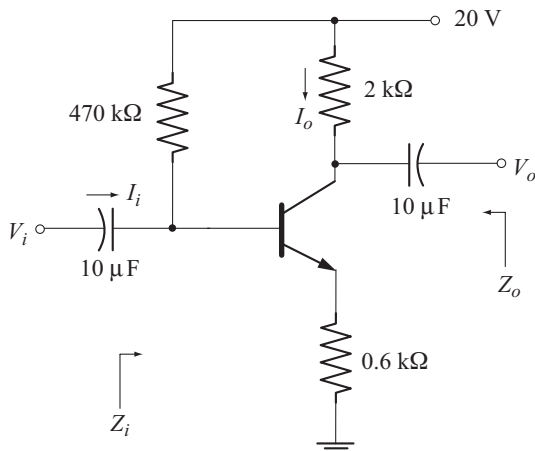
Fig. 3.49 AC equivalent circuit

Example 3.9

For the circuit shown below calculate

- (a) r_e (b) Z_i
 (c) Z_o (d) A_v
 (e) A_i

Take $\beta = 120$ and $r_o = 40 \text{ k}\Omega$

**Solution**

$$\begin{aligned}
 \text{(a)} \quad I_B &= \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta) R_E} \\
 &= \frac{20 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (121)(0.6 \text{ k}\Omega)} = 35.57 \mu\text{A} \\
 I_E &= (1 + \beta) I_B = (121)(35.57 \mu\text{A}) = 4.3 \text{ mA}
 \end{aligned}$$

$$r_e = \frac{26\text{mV}}{I_E} = \frac{26\text{mV}}{4.3\text{mA}} = 6.04\ \Omega$$

$$(b) \quad Z_b = \beta (R_E + r_e) = 120 (600\ \Omega + 6.04\ \Omega) = 72.72\ \text{k}\Omega$$

Now

$$Z_i = Z_b \parallel R_B = 72.72\ \text{k}\Omega \parallel 470\ \text{k}\Omega = 62.98\ \text{k}\Omega$$

$$(c) \quad Z_o \approx R_C = 2\ \text{k}\Omega$$

$$(d) \quad A_V \approx - \frac{\beta R_C}{Z_b} = - \frac{(120)(2\ \text{k}\Omega)}{(72.72\ \text{k}\Omega)} = -3.3$$

$$(e) \quad A_I = - \frac{A_V Z_i}{R_C} = - \frac{(-3.3)(62.98\ \text{k}\Omega)}{2\ \text{k}\Omega} = 103.92.$$

Note : If R_E is bypassed by C_E

- r_e remain unchanged since C_E does not affect dc conditions.
- In the calculations of Z_b , Z_i , A_V and A_I , we have to substitute $R_E = 0\ \Omega$, since R_E will be shorted out by C_E .

◆ 3.22 COMMON-EMITTER CONFIGURATION USING VOLTAGE DIVIDER BIAS WITH UNBYPASSED R_E

Figure 3.50 shows common-emitter configuration using voltage divider bias with unbypassed R_E . The ac equivalent circuit is shown in Fig. 3.51. Note that this ac equivalent circuit is exactly identical to that given in Fig. 3.44 with R_B equal to the parallel combination of R_1 and R_2 . Hence the analysis given in section 3.20 can be readily applied to this circuit.

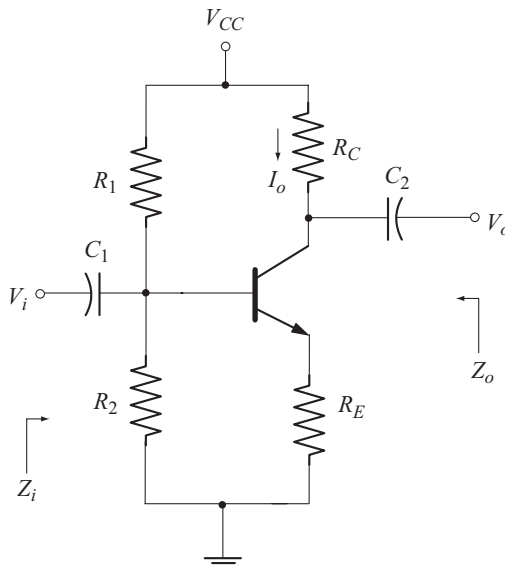


Fig. 3.50 CE Voltage divider configuration with un bypassed R_E

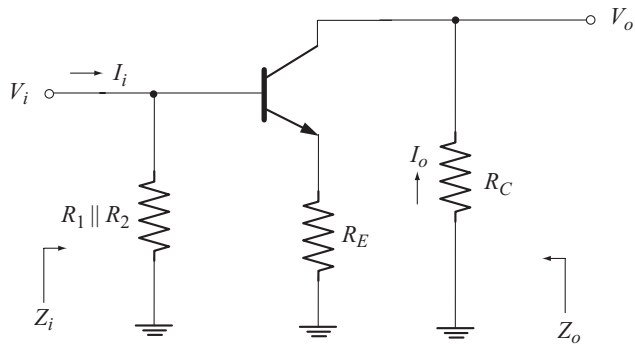


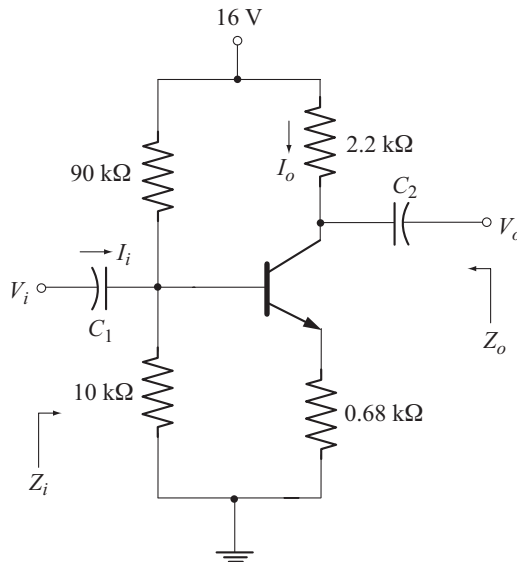
Fig. 3.51 AC equivalent circuit

Example 3.10

For the circuit shown below calculate

- (a) r_e , Z_i and Z_o (b) A_v and A_i

Take $\beta = 210$ and $r_o = 50 \text{ k}\Omega$.



Solution

Using approximate analysis for voltage divider bias (since $\beta R_E > 10 R_2$) we get

$$I_E = 1.324 \text{ mA}$$

(a)
$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.324 \text{ mA}} = 19.64 \Omega.$$

$$R_B = R_{Th} = R'$$

$$= R_1 \parallel R_2 = 90 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 9 \text{ k}\Omega$$

$$Z_b \approx \beta(r_e + R_E) = 210 (19.64 \Omega + 0.68 \text{ k}\Omega) = 146.92 \text{ k}\Omega$$

$$Z_i = Z_b \parallel R_B = 146.92 \text{ k}\Omega \parallel 9 \text{ k}\Omega = 8.48 \text{ k}\Omega$$

$$Z_o = R_C = 2.2 \text{ k}\Omega$$

$$(b) \quad A_V \approx - \frac{R_C}{r_e + R_E} = - \frac{2.2 \text{ k}\Omega}{19.64 \Omega + 0.68 \text{ k}\Omega} = -3.14$$

$$A_I = - \frac{A_V Z_i}{R_C} = - \frac{(-3.14)(8.48 \text{ k}\Omega)}{(2.2 \text{ k}\Omega)} = 12.1$$

◆ 3.23 VOLTAGE DIVIDER BIAS AND EMITTER BIAS CONFIGURATIONS WITH PARTLY BYPASSED R_E

Figure 3.52 shows emitter-bias configuration with partly bypassed R_E . The voltage divider bias configuration with partly bypassed R_E is shown in Fig. 3.53.

For dc operation, C_E acts as an open circuit. Hence the emitter resistance for dc analysis (i.e., for the calculation of r_e) is $R_E = R_{E1} + R_{E2}$.

For ac operation, R_{E2} is shorted out by C_E . Hence for ac analysis (i.e., to calculate Z_i , A_V etc) the emitter resistance is, $R_E = R_{E1}$.

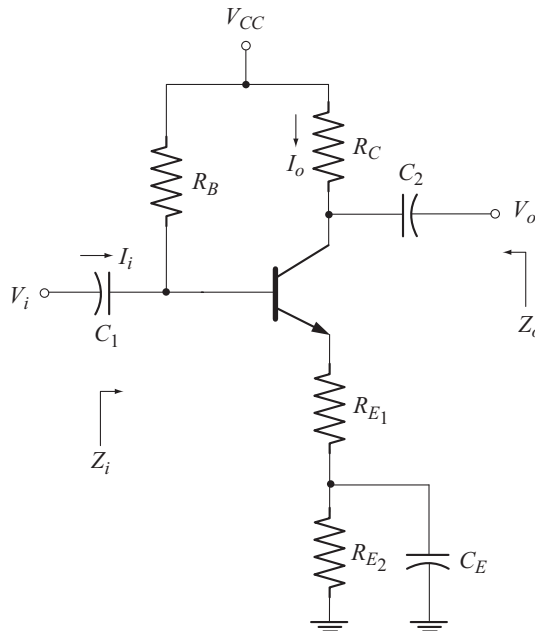


Fig. 3.52 Emitter-bias configuration with partly bypassed R_E

For both the circuits, analysis given in section 3.20 can be used. It is important to note that, for voltage divider configuration we have to take

$$R_B = R_{Th} = R' = R_1 \parallel R_2$$

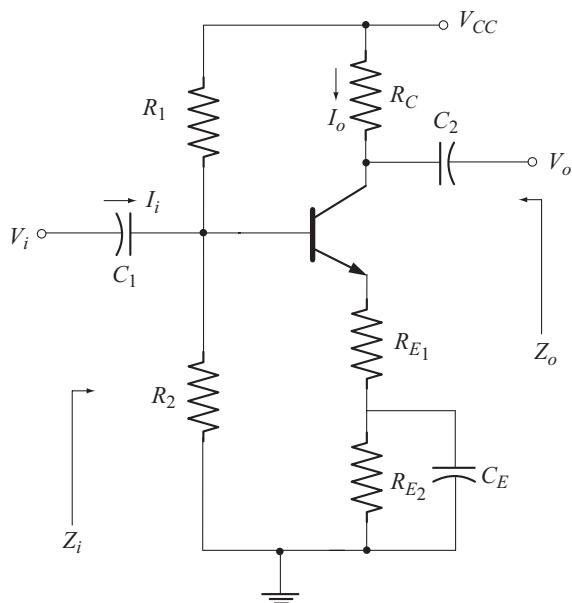


Fig. 3.53 Voltage divider-bias configuration with partly bypassed R_E

◆ 3.24 EMITTER-FOLLOWER CONFIGURATION

Figure 3.54 shows the circuit of emitter-follower configuration.

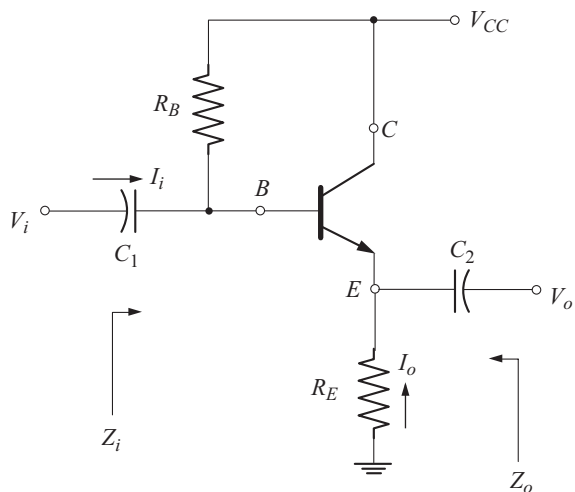


Fig. 3.54 Emitter-follower configuration

The input signal V_i is applied to the base of transistor through the input coupling capacitor C_1 . The output signal V_o is taken at the emitter through the output coupling capacitor C_2 .

Let us draw the ac equivalent circuit by reducing V_{CC} to zero and replacing the capacitors C_1 and C_2 by their short circuit equivalents as shown in Fig. 3.55.

Observe that the collector terminal is common to input and output circuits. Hence the configuration is common collector. The ac input voltage V_i is the sum of ac base-emitter voltage V_{be} and ac output voltage V_o .

$$\text{ie.} \quad V_i = V_{be} + V_o$$

Neglecting V_{be} we can write

$$V_o \approx V_i \quad (3.82)$$

The output voltage (emitter voltage) follows the input voltage (base voltage). Hence the name emitter follower.

From Equation (3.82) we have

$$A_V = \frac{V_o}{V_i} \approx 1 \quad (3.83)$$

Thus for emitter follower the voltage gain is approximately unity.

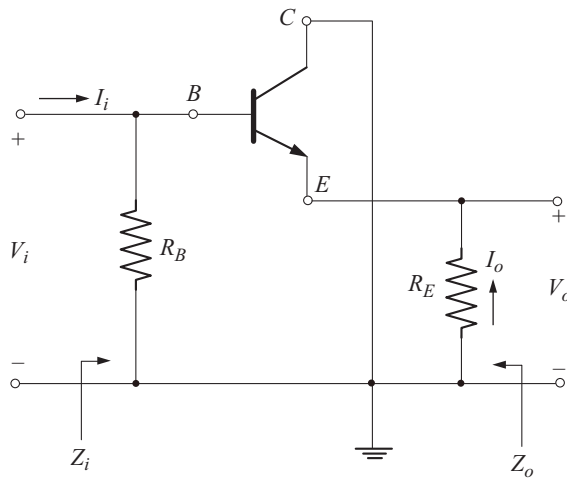


Fig. 3.55 AC equivalent circuit

Let us replace the transistor in the circuit of Fig. 3.55 by its r_e model as shown in Fig. 3.56. Note that r_o is not considered for simplicity.

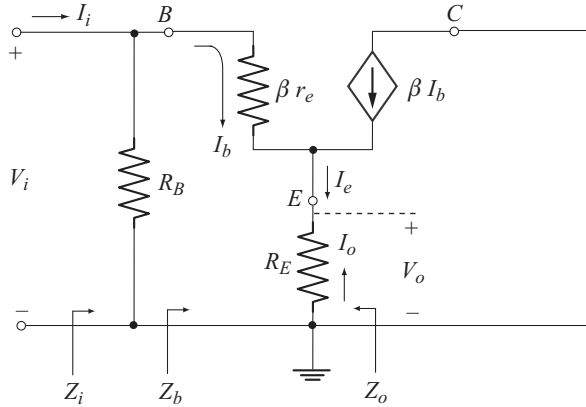


Fig. 3.56 AC equivalent circuit with r_e model

Input Impedance (Z_i)

Applying KVL to the input circuit we have

$$V_i = I_b \beta r_e + I_e R_E$$

Using $I_e = (1 + \beta) I_b$ we have

$$V_i = I_b \beta r_e + (1 + \beta) I_b R_E$$

$$\text{Now } Z_b = \frac{V_i}{I_b} = \beta r_e + (1 + \beta) R_E \quad (3.84)$$

$$Z_i = \frac{V_i}{I_i} = R_B \parallel Z_b \quad (3.85)$$

Since $(1 + \beta) \approx \beta$, Equation (3.84) becomes

$$Z_b \approx \beta(r_e + R_E) \quad (3.86)$$

Taking $r_e + R_E \approx R_E$, we can write

$$Z_b \approx \beta R_E \quad (3.87)$$

Output Impedance (Z_o)

Consider

$$Z_b = \frac{V_i}{I_b} \Rightarrow I_b = \frac{V_i}{Z_b}$$

Using

$$I_b = \frac{I_e}{1 + \beta} \quad \text{we have}$$

$$\frac{I_e}{1 + \beta} = \frac{V_i}{Z_b}$$

$$I_e = \frac{(1 + \beta) V_i}{Z_b}$$

Substituting for Z_b from Equation (3.84) we have

$$I_e = \frac{(1+\beta)V_i}{\beta r_e + (1+\beta)R_E} \quad (3.88)$$

$$(1 + \beta) \approx \beta$$

$$\beta r_e + (1 + \beta) R_E \approx \beta (r_e + R_E)$$

Using these relations in Equation (3.88) we have

$$I_e \approx \frac{\beta V_i}{\beta (r_e + R_E)}$$

$$V_i \approx I_e (r_e + R_E)$$

$$\text{or} \quad V_i \approx I_e r_e + I_e R_E \quad (3.89)$$

Equation (3.89) is KVL equation and its circuit representation is shown in Fig. 3.57.

From Fig. 3.57 we note that $I_e R_E$ gives the output voltage V_o .

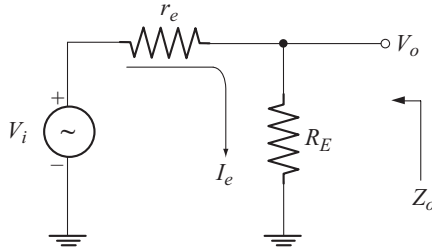


Fig. 3.57 Circuit representation of Equation (3.89)

To find Z_o let us set V_i to zero. The resulting circuit is shown in Fig. 3.58.

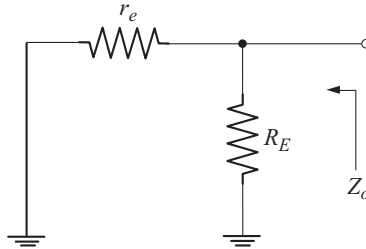


Fig. 3.58 Circuit to find Z_o

Since r_e and R_E are in parallel we have

$$Z_o = r_e \parallel R_E = \frac{r_e R_E}{r_e + R_E} \quad (3.90)$$

Taking $r_e + R_E \approx R_E$, we get

$$Z_o \approx \frac{r_e R_E}{R_E}$$

or $Z_o \approx r_e$ (3.91)

Voltage Gain (A_V)

Applying voltage division rule to the circuit of Fig. 3.57 we have

$$V_o = \frac{R_E}{r_e + R_E} V_i$$

Now $A_V = \frac{V_o}{V_i} = \frac{R_E}{r_e + R_E}$ (3.92)

Usually $R_E \gg r_e \quad \therefore \quad R_E + r_e \approx R_E$

$$A_V = \frac{V_o}{V_i} \approx 1$$
 (3.93)

or $V_o \approx V_i$

Note that the output voltage follows the input voltage as stated earlier.

Current Gain (A_I)

$$V_o = I_e R_E = -I_o R_E \quad [\because I_e = -I_o]$$

$$\therefore I_o = -\frac{V_o}{R_E}$$

Also $I_i = \frac{V_i}{Z_i}$

Now $A_I = \frac{I_o}{I_i}$

$$= \left[-\frac{V_o}{R_E} \right] \div \left[\frac{V_i}{Z_i} \right]$$

$$= - \left[\frac{V_o}{V_i} \right] \left[\frac{Z_i}{R_E} \right]$$

$$A_I = - \frac{A_V Z_i}{R_E}$$
 (3.94)

Phase Relationship

The positive sign for A_V in Equation (3.93) reveals that V_o and V_i are in phase.

Note : r_e has no significant effect on ac analysis.

3.24.1 Important Characteristics of Emitter Follower

Based on the results obtained, we can list the following important characteristics of emitter follower.

- The input impedance is high.
- The output impedance is low.
- The voltage gain is approximately unity.
- The input and output voltages are in phase.
- The current gain is high.

Since emitter follower presents a high impedance at the input and a low impedance at the output it is used for impedance matching purposes. It is used as a buffer between the voltage source of high source impedance and a low impedance load.

◆ 3.25 EMITTER FOLLOWER USING VOLTAGE DIVIDER BIAS

Figure 3.59 shows an emitter follower using voltage divider bias. The ac analysis given in section 3.24 can also be used for this circuit, simply by replacing R_B by $R_1 \parallel R_2$.

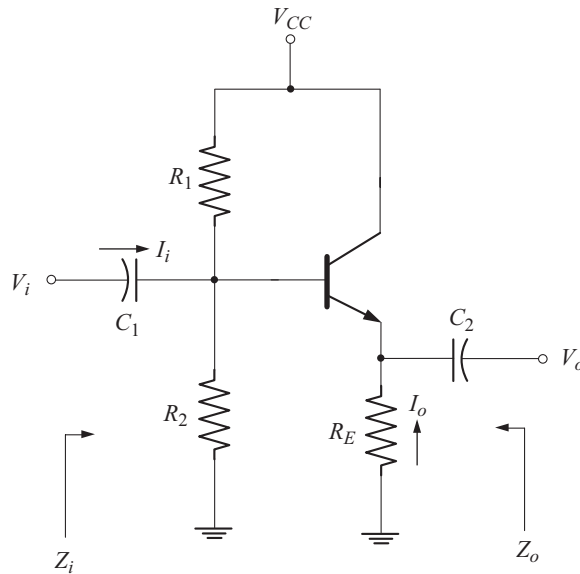


Fig. 3.59 Emitter follower using voltage divider bias

◆ 3.26 EMITTER FOLLOWER USING COLLECTOR RESISTOR

Figure 3.60 shows the circuit of emitter follower using collector resistor R_C . The presence of R_C affects only the dc operation of the circuit. Since R_C is not reflected into the base or emitter equivalent networks the ac analysis given in section 3.24 can be applied to this circuit simply by replacing R_B by $R_1 \parallel R_2$. It is important to note that since the addition of R_C does not affect I_E , the value of r_e remains unchanged. Only V_{CE} will get affected.